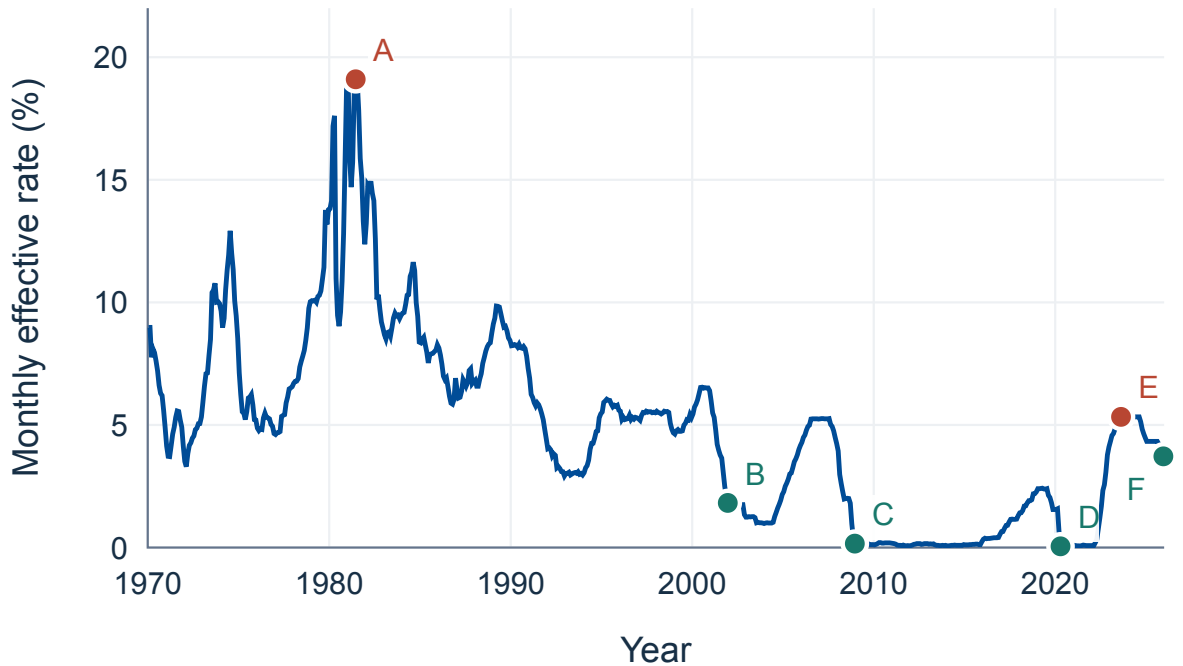
Oxford College of Emory University

Chapter 12

Aggregate Demand and Aggregate Supply

Guilherme Paiva Pinto · ECON_OX 112

The Federal Funds Rate, 1970–2025



Data: Federal Reserve Board H.15 / FEDFUNDS via FRED.
Monthly effective rates; January 1970–December 2025.

Context: Fed History · 2001 · 2008 · 2020 · 2022 · 2024 Board statements.

The Federal Funds Rate, 1970–2025



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Monthly effective rates; January 1970–December 2025.

Rates rising

A · 1979–1981
Persistent inflation; tighter policy.

E · 2022–2023
High inflation; rate increases.

Context: Fed History · 2001 · 2008 · 2020 · 2022 · 2024 Board statements.

The Federal Funds Rate, 1970–2025



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Monthly effective rates; January 1970–December 2025.

Rates rising

A · 1979–1981
Persistent inflation; tighter policy.

E · 2022–2023
High inflation; rate increases.

Rates falling

B · 2001 · Weak demand.

C · 2007–2008 · Credit crisis.

D · 2020 · Pandemic disruption.

F · 2024–2025 · Easing inflation.

Context: Fed History · 2001 · 2008 · 2020 · 2022 · 2024 Board statements.

The Federal Reserve

Goals

Maximum employment and stable prices: 2% inflation over time.

Source: Federal Reserve Board: monetary-policy goals

The Federal Reserve

Goals

Maximum employment and stable prices: 2% inflation over time.

Policy tool

The federal funds rate target influences borrowing costs.

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The Federal Reserve

Goals

Maximum employment and stable prices: 2% inflation over time.

Policy tool

The federal funds rate target influences borrowing costs.

Spending response

Higher rates restrain spending; lower rates support spending.

AD–AS connects spending with real GDP and the price level.

Source: Federal Reserve Board: monetary-policy goals

Roadmap

Aggregate demand

Spending plans and demand shifts

Aggregate supply

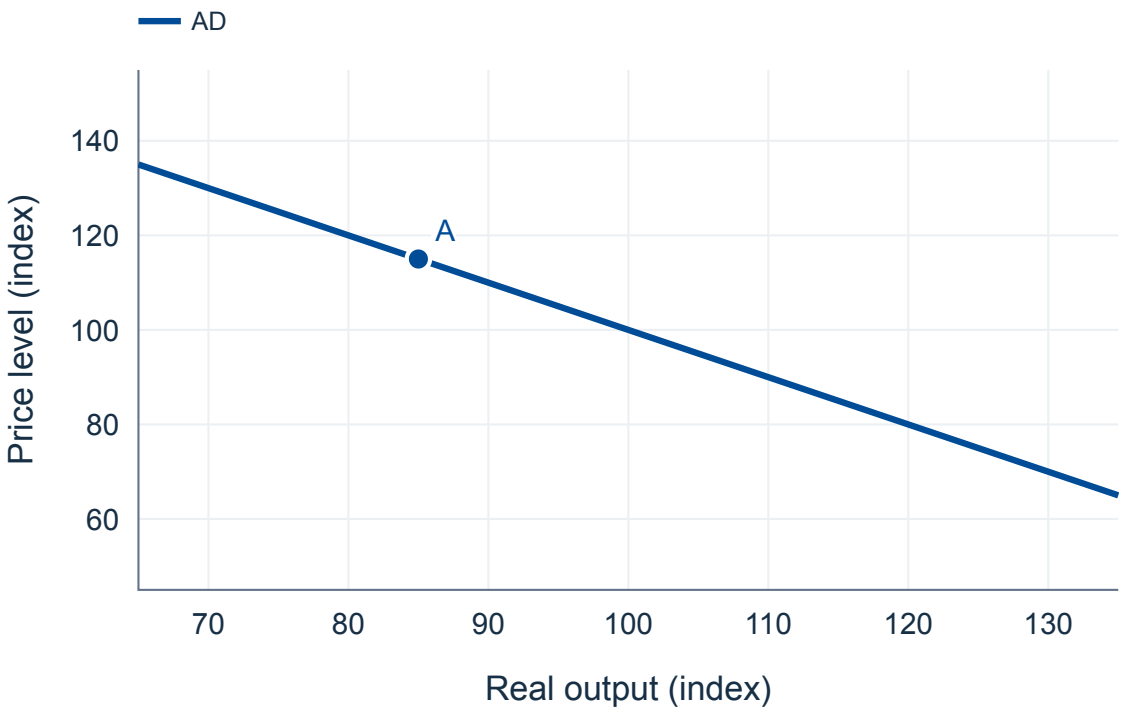
Costs, sticky wages, and productive capacity

Equilibrium and adjustment

Output gaps, shocks, and worked graphical cases

Aggregate Demand

Y = C + I + G + NX

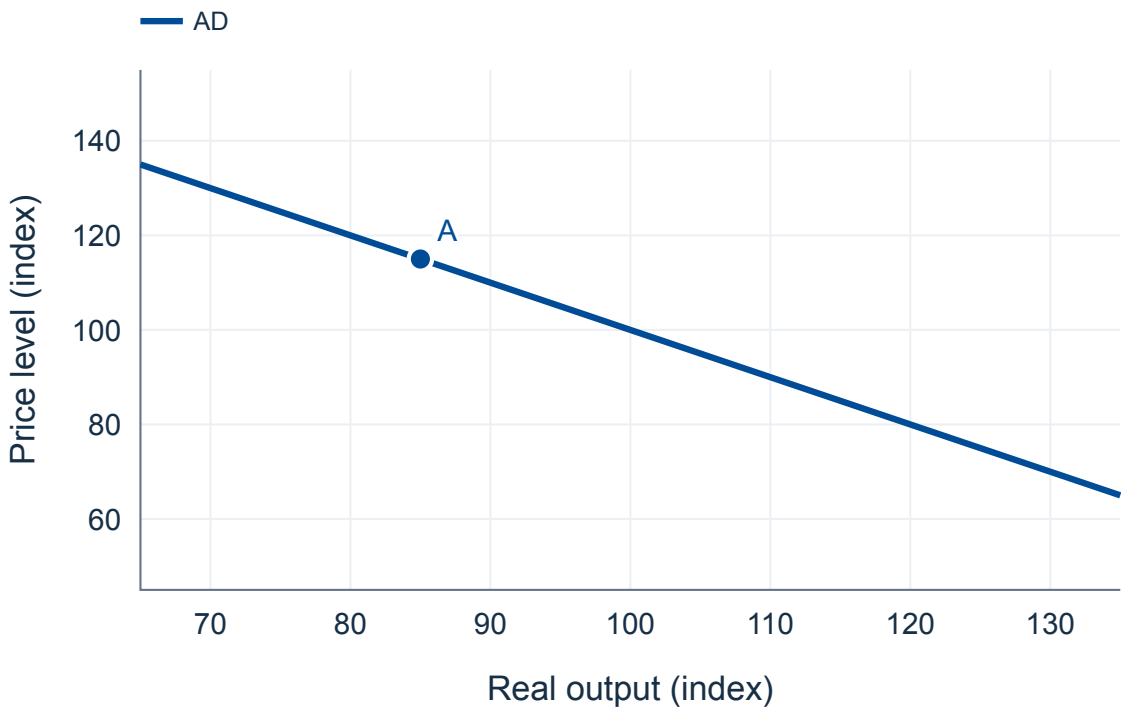


Definition

Planned spending on domestic final output at each price level.

Aggregate Demand

Y = C + I + G + NX



Definition

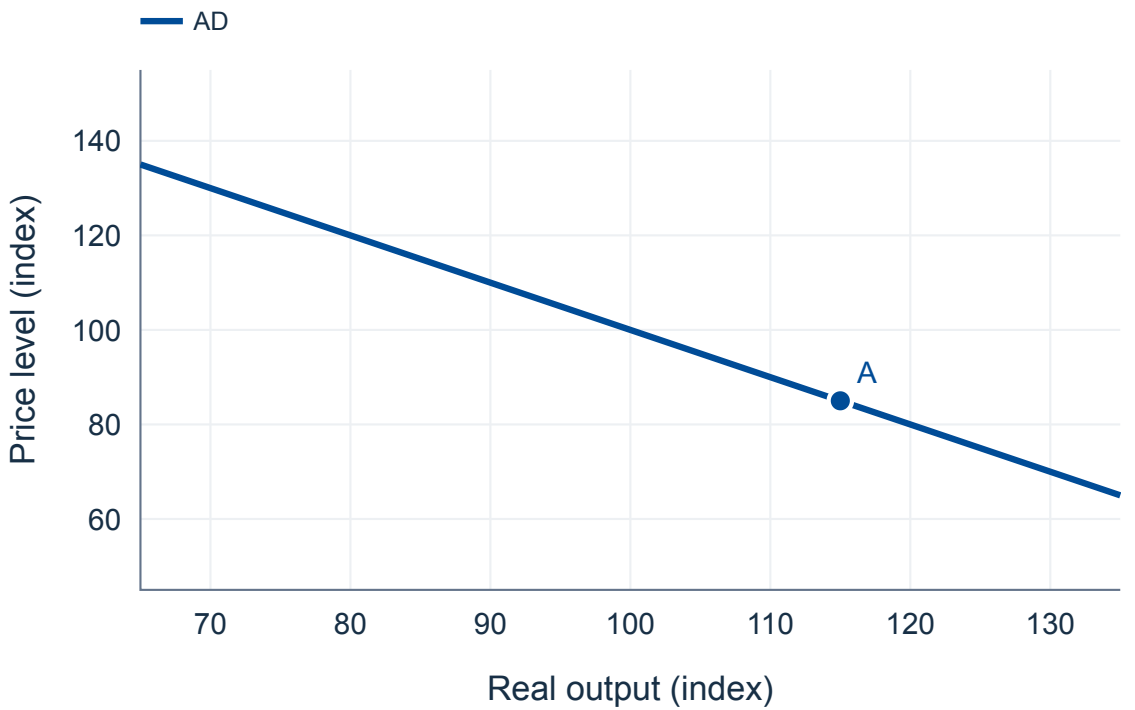
Planned spending on domestic final output at each price level.

Downward slope

A lower price level can support consumption and investment.

Aggregate Demand

Y = C + I + G + NX



Definition

Planned spending on domestic final output at each price level.

Downward slope

A lower price level can support consumption and investment.

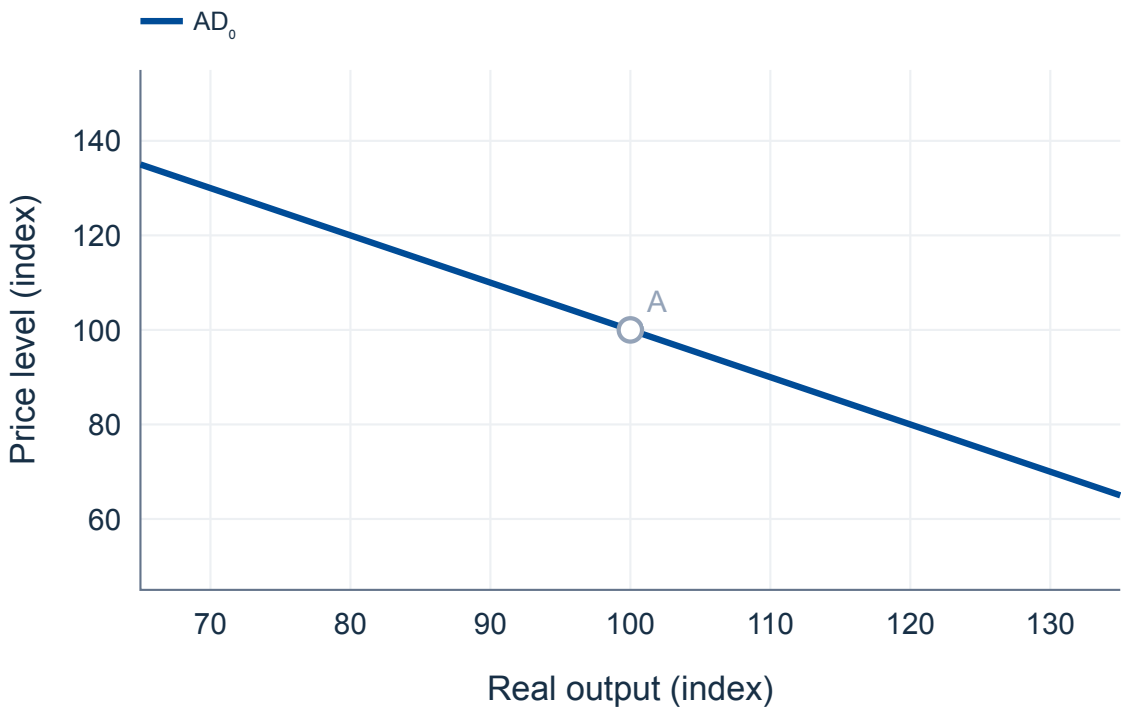
Movement along AD

Other demand influences are held fixed.

The axes measure real GDP and the overall price level.

Movements Along AD and Shifts

Compare one change at a time

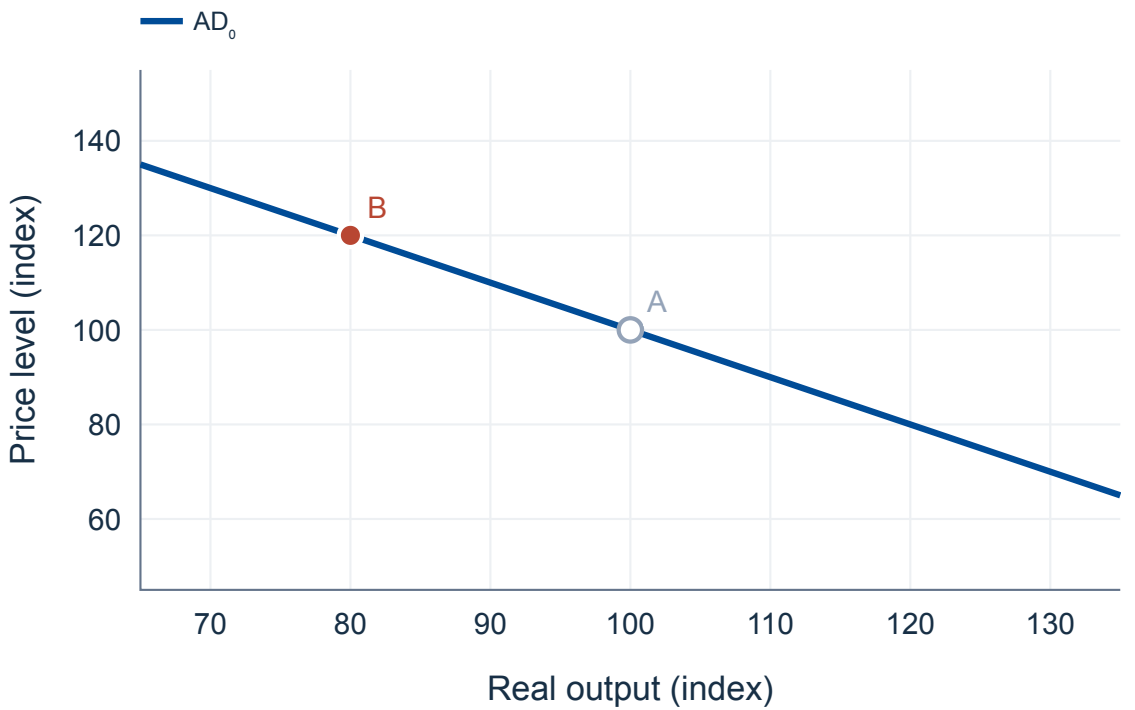


Starting point A

At a price level of 100, desired output is 100.

Movements Along AD and Shifts

Compare one change at a time



Starting point A

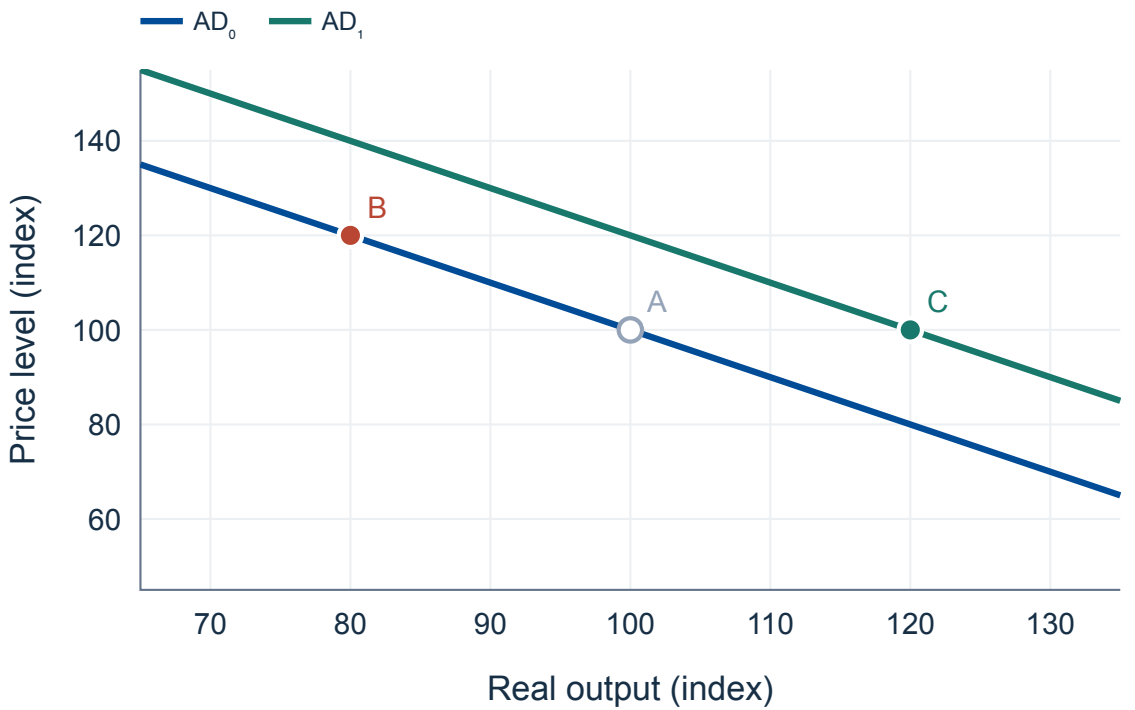
At a price level of 100, desired output is 100.

Along the same curve

Higher prices reduce desired output: A to B.

Movements Along AD and Shifts

Compare one change at a time



Starting point A

At a price level of 100, desired output is 100.

Along the same curve

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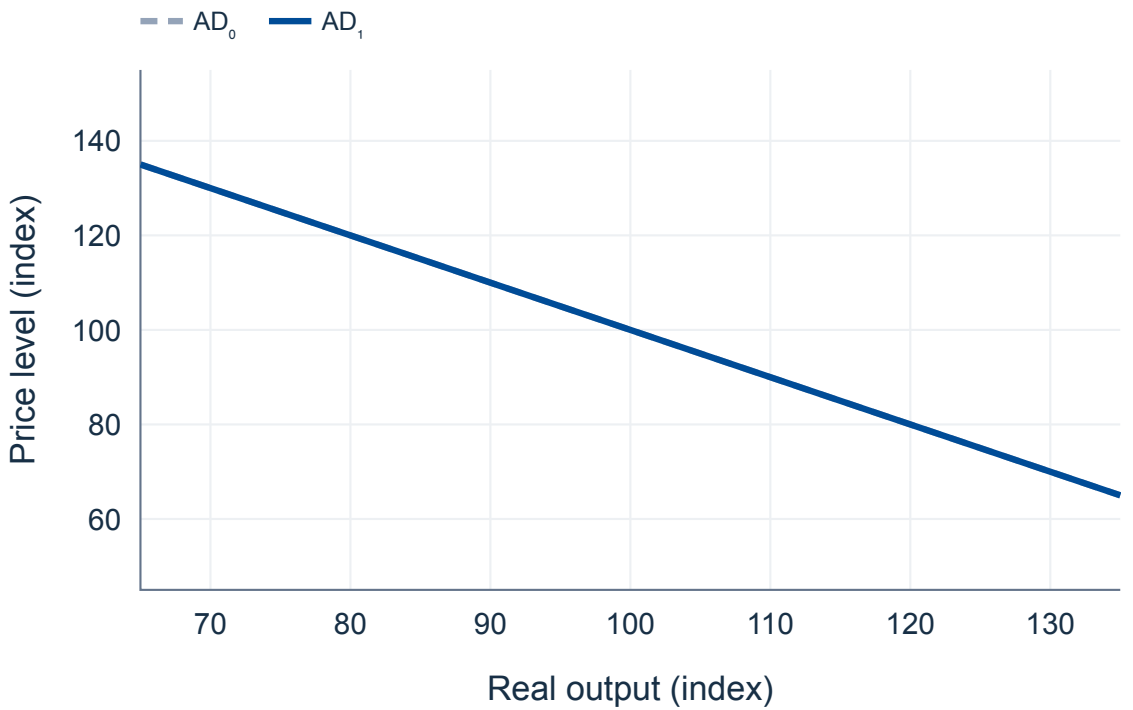
A shift of the curve

Greater confidence raises spending at the same price level: A to C.

Price-level change: movement. Other demand change: shift.

Factors That Shift Aggregate Demand

At the same price level · Other determinants unchanged

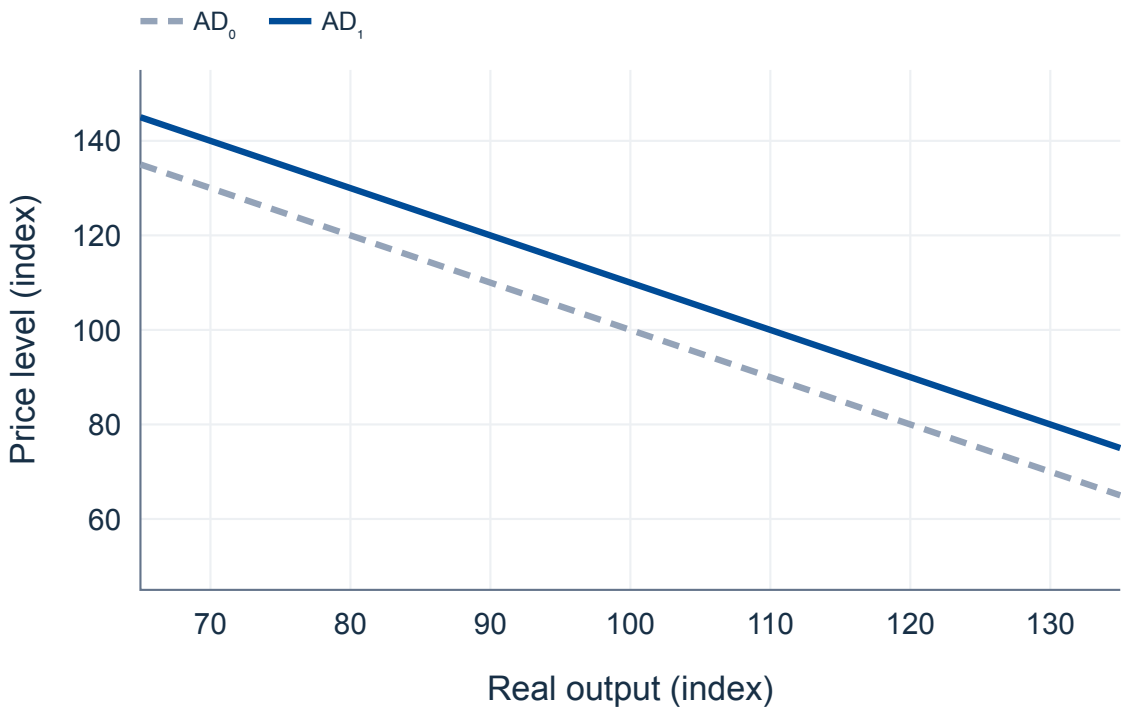


Households

Higher expected income or wealth can raise C.

Factors That Shift Aggregate Demand

At the same price level · Other determinants unchanged



Households

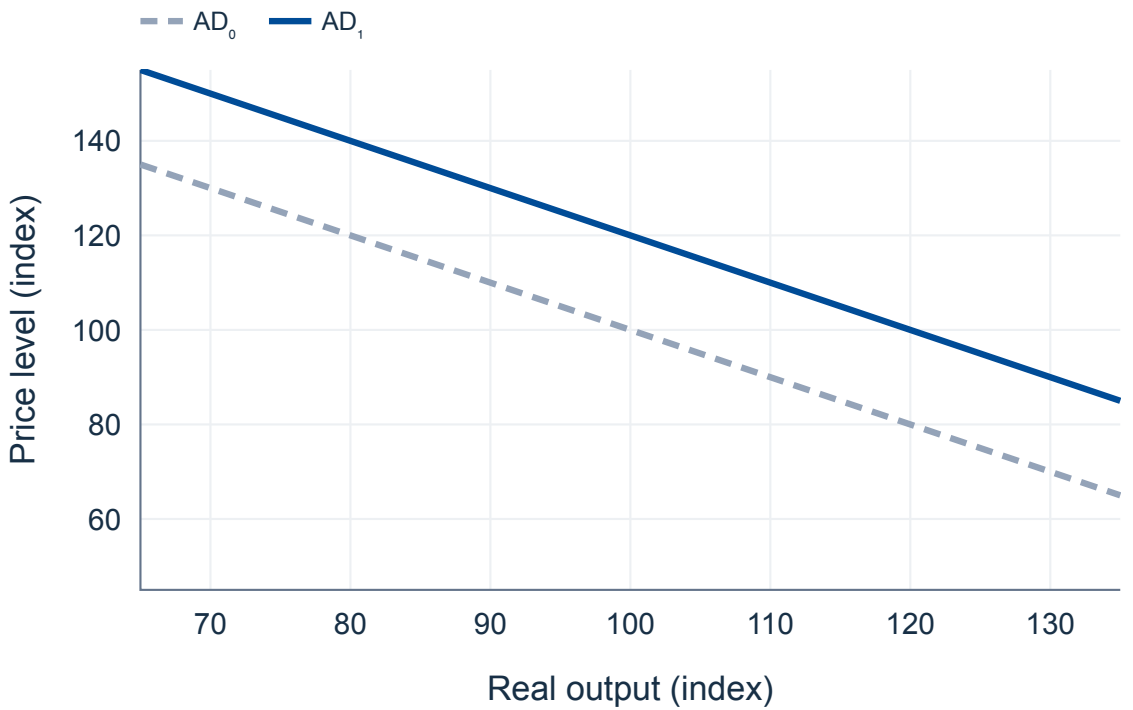
Higher expected income or wealth can raise C.

Firms

Higher expected sales can raise I; idle capital can reduce I.

Factors That Shift Aggregate Demand

At the same price level · Other determinants unchanged



Households

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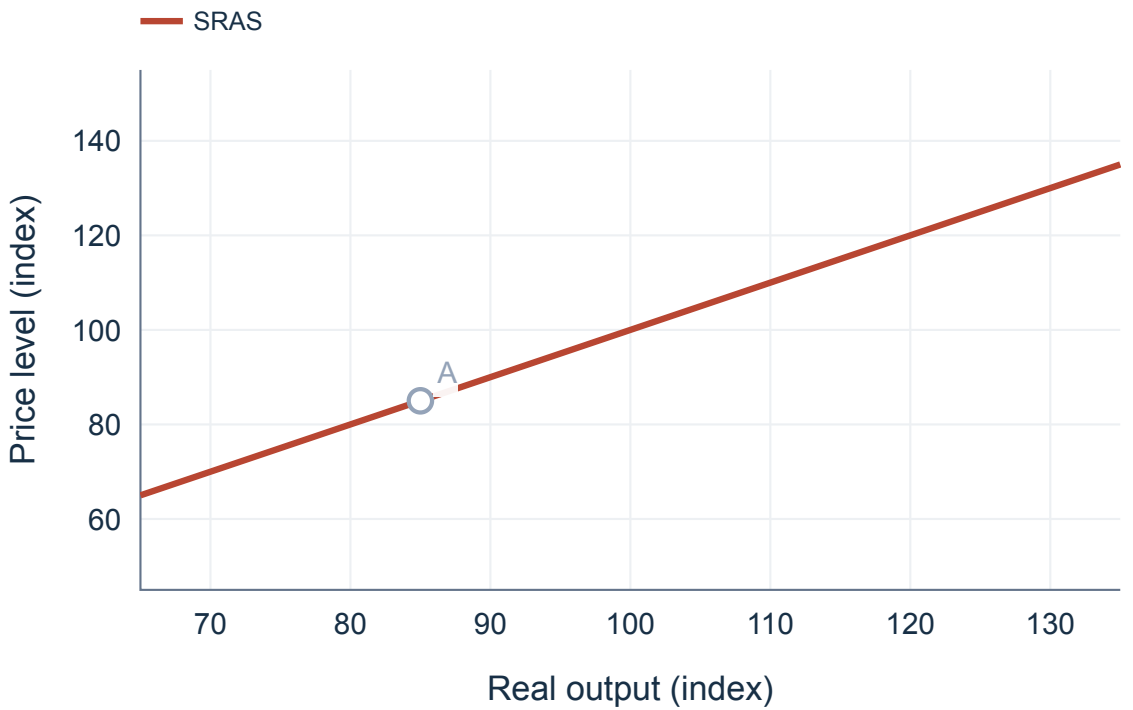
Policy

Higher G, lower taxes, or lower interest rates can support AD.

Higher planned C, I, G, or NX shifts AD right.

Short-Run Aggregate Supply

A bakery’s wage contract fixes hourly pay at $20 for the month

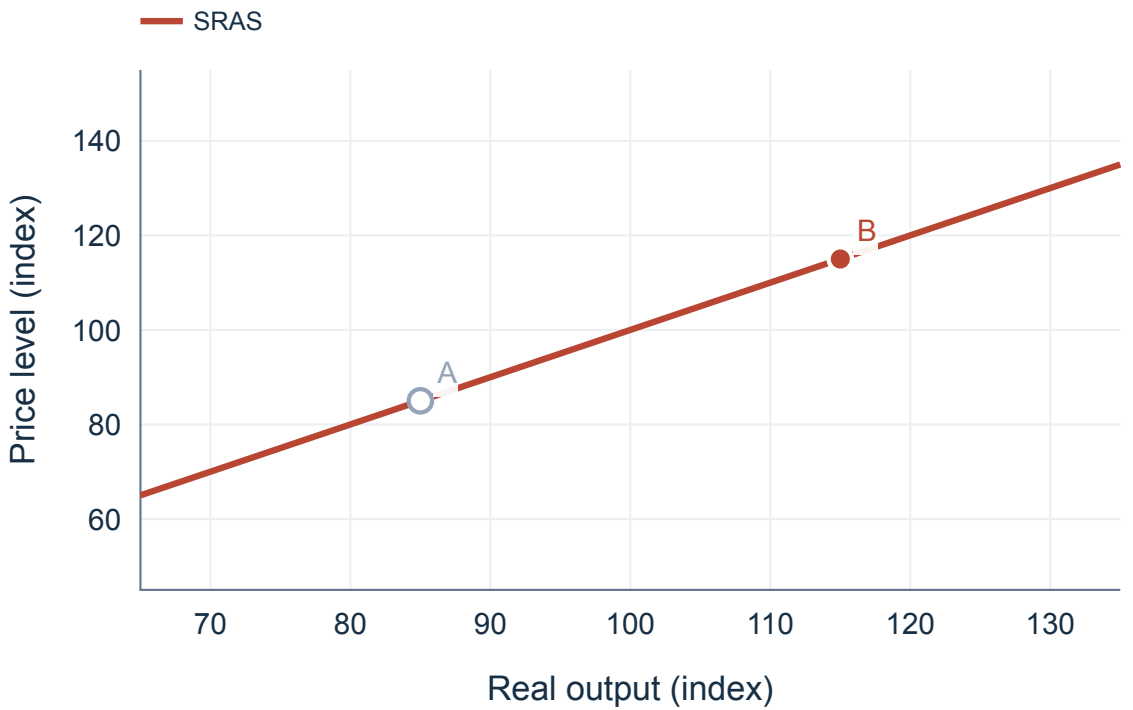


Some input costs adjust slowly

Nominal pay remains fixed temporarily.

Short-Run Aggregate Supply

A bakery’s wage contract fixes hourly pay at $20 for the month



Some input costs adjust slowly

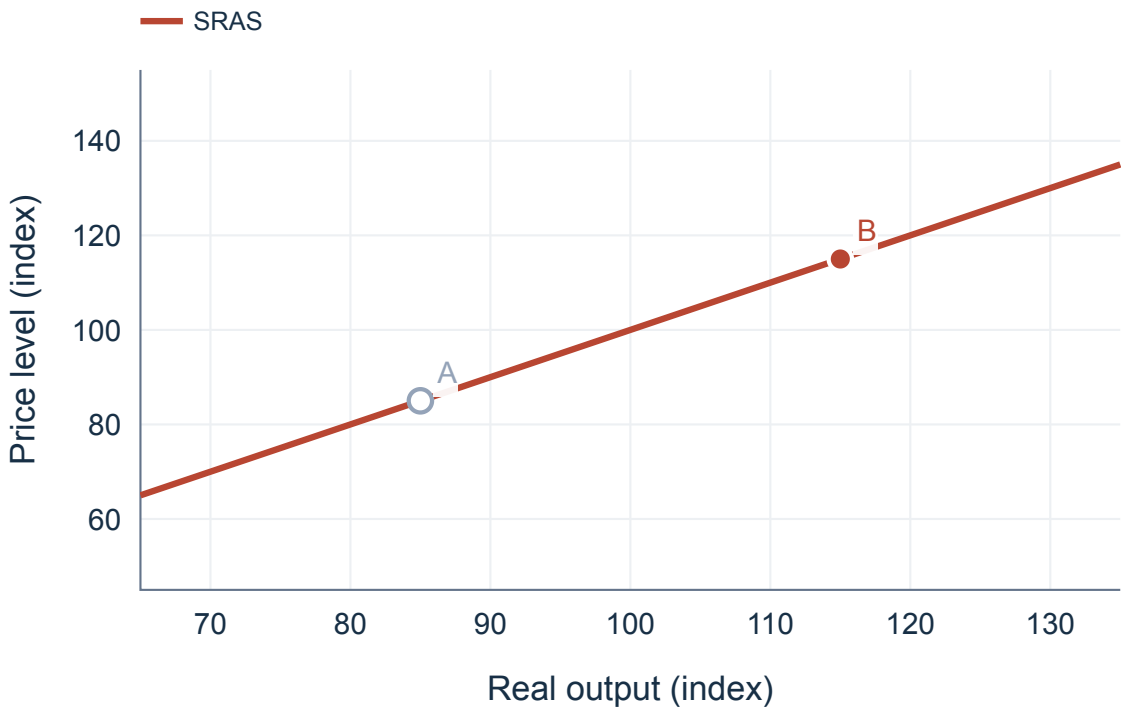
Nominal pay remains fixed temporarily.

Selling prices rise first

Extra production can become more profitable.

Short-Run Aggregate Supply

A bakery's wage contract fixes hourly pay at $20 for the month



Some input costs adjust slowly

Nominal pay remains fixed temporarily.

Selling prices rise first

Extra production can become more profitable.

Output increases

The economy moves along an upward-sloping SRAS curve.

Sticky wages mean delayed adjustment.

Sticky Prices and Short-Run Output

> **Adjustment takes time**
>
> Contracts and repricing costs delay changes in wages and prices.

Source: Federal Reserve Board: price and wage adjustment

Sticky Prices and Short-Run Output

Adjustment takes time

Contracts and repricing costs delay changes in wages and prices.

Fewer orders

Firms may cut production and hiring before changing prices.

Source: Federal Reserve Board: price and wage adjustment

Sticky Prices and Short-Run Output

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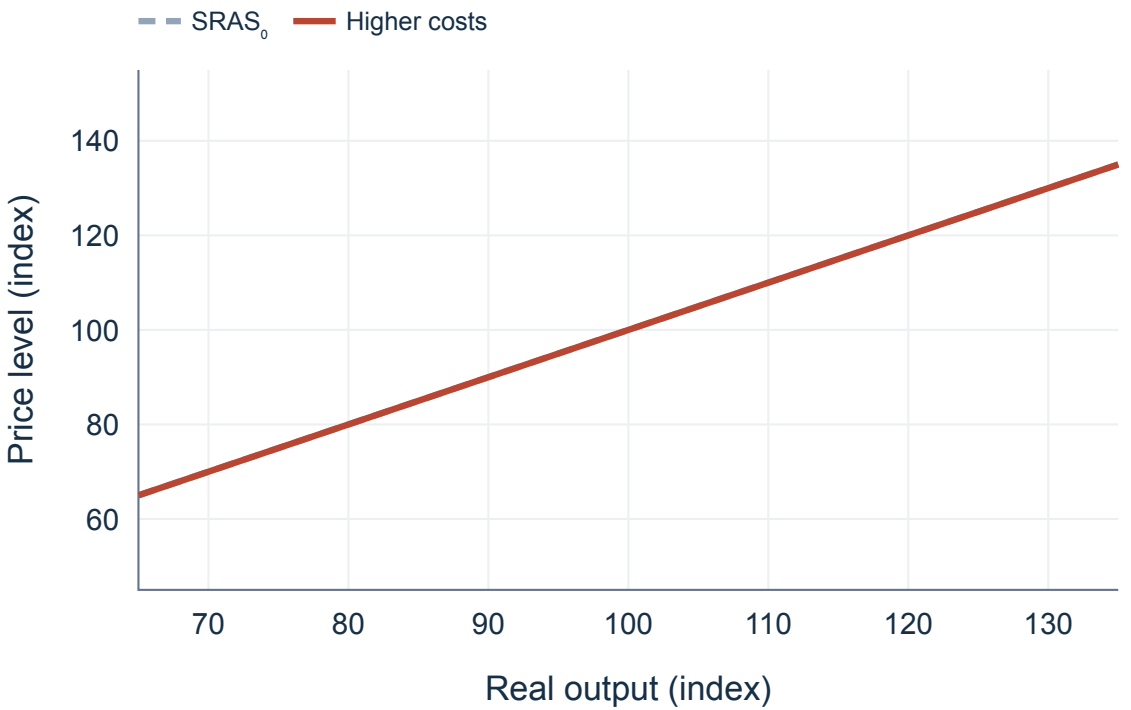
Short-run consequence

Spending shocks can change real output as well as prices.

Source: Federal Reserve Board: price and wage adjustment

Factors That Shift SRAS

Hold the overall price level fixed to identify a shift

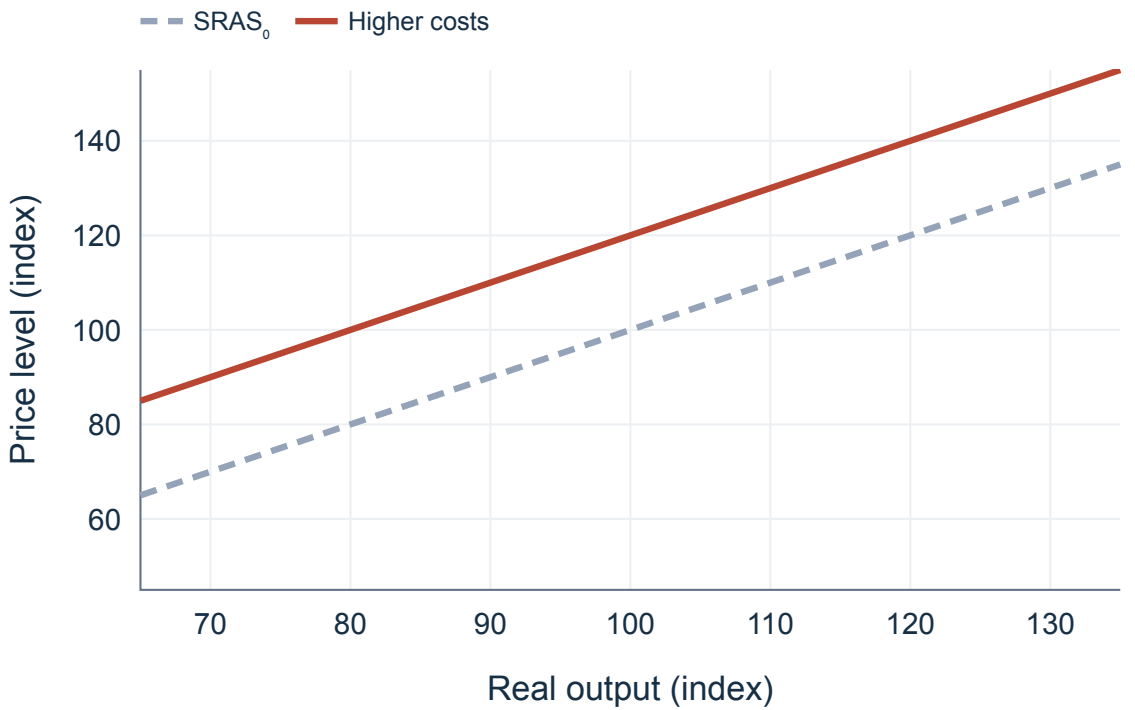


Higher input costs

More expensive energy shifts SRAS left.

Factors That Shift SRAS

Hold the overall price level fixed to identify a shift



Higher input costs

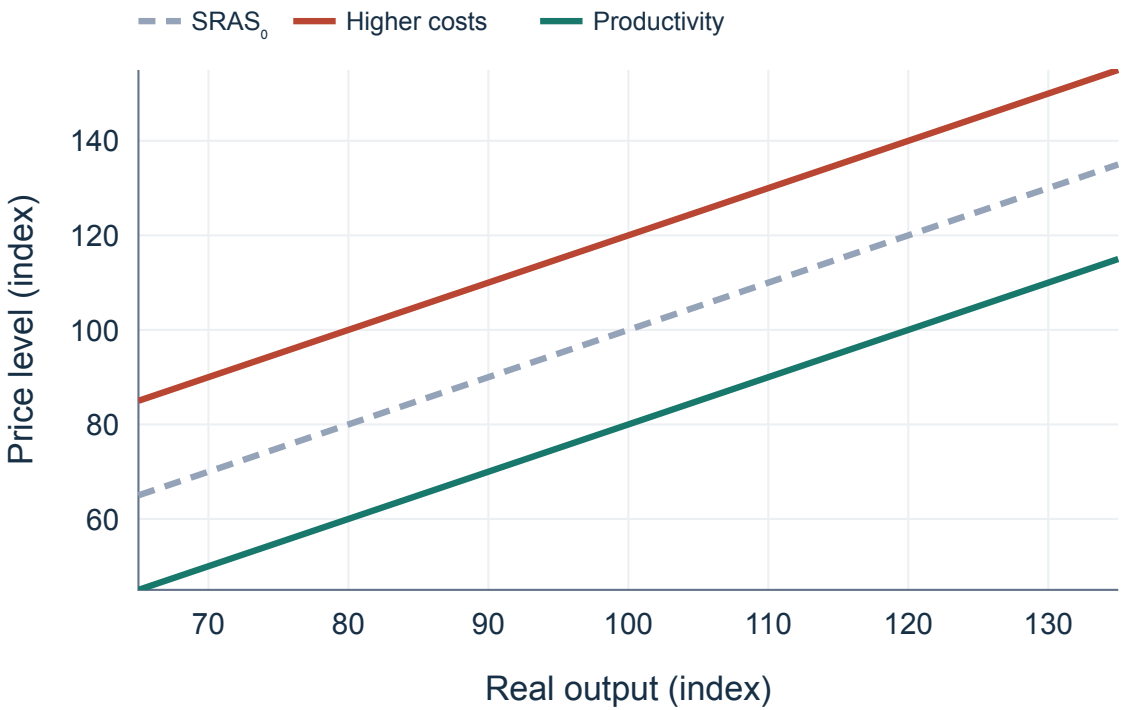
More expensive energy shifts SRAS left.

Higher expected prices

Higher agreed nominal wages can raise costs and shift SRAS left.

Factors That Shift SRAS

Hold the overall price level fixed to identify a shift



Higher input costs

More expensive energy shifts SRAS left.

Higher expected prices

Higher agreed nominal wages can raise costs and shift SRAS left.

Higher productivity

Lower unit costs shift SRAS right.

A change in the price level itself moves the economy along SRAS.

AD and SRAS Checkpoint

Name the curve, its direction, and the reason. Hold other determinants fixed.

Households plan to spend less at every price level.

Energy used in production becomes cheaper.

AD and SRAS Checkpoint

Name the curve, its direction, and the reason. Hold other determinants fixed.

Households plan to spend less at every price level.

AD shifts left: planned consumption falls.

Energy used in production becomes cheaper.

AD and SRAS Checkpoint

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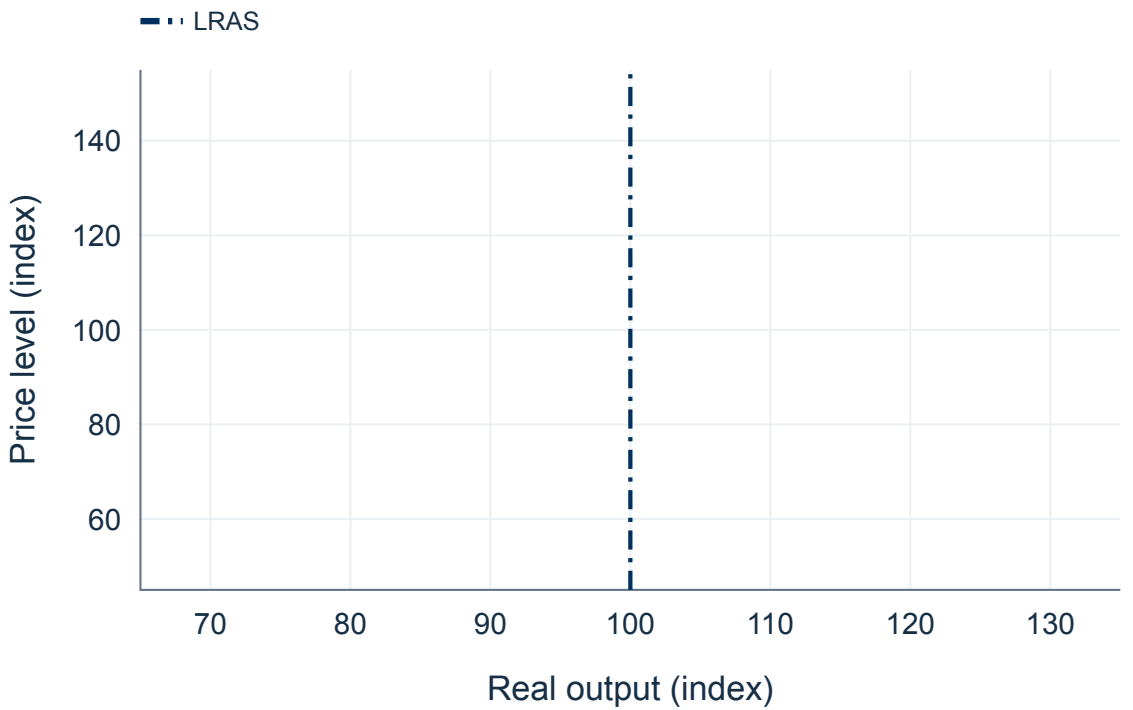
AD shifts left: planned consumption falls.

Energy used in production becomes cheaper.

SRAS shifts right: production costs fall.

Long-Run Aggregate Supply

Potential output = 100 · Output and price level are indices

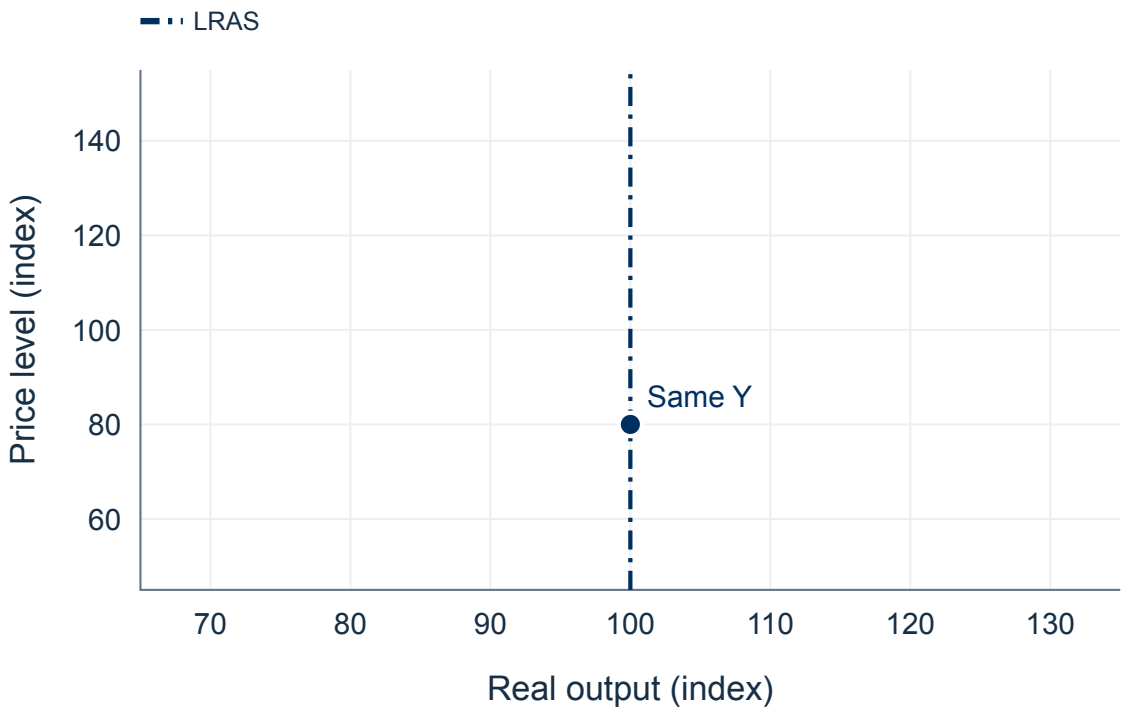


Potential output

Sustainable production at normal resource-use rates.

Long-Run Aggregate Supply

Potential output = 100 · Output and price level are indices



Potential output

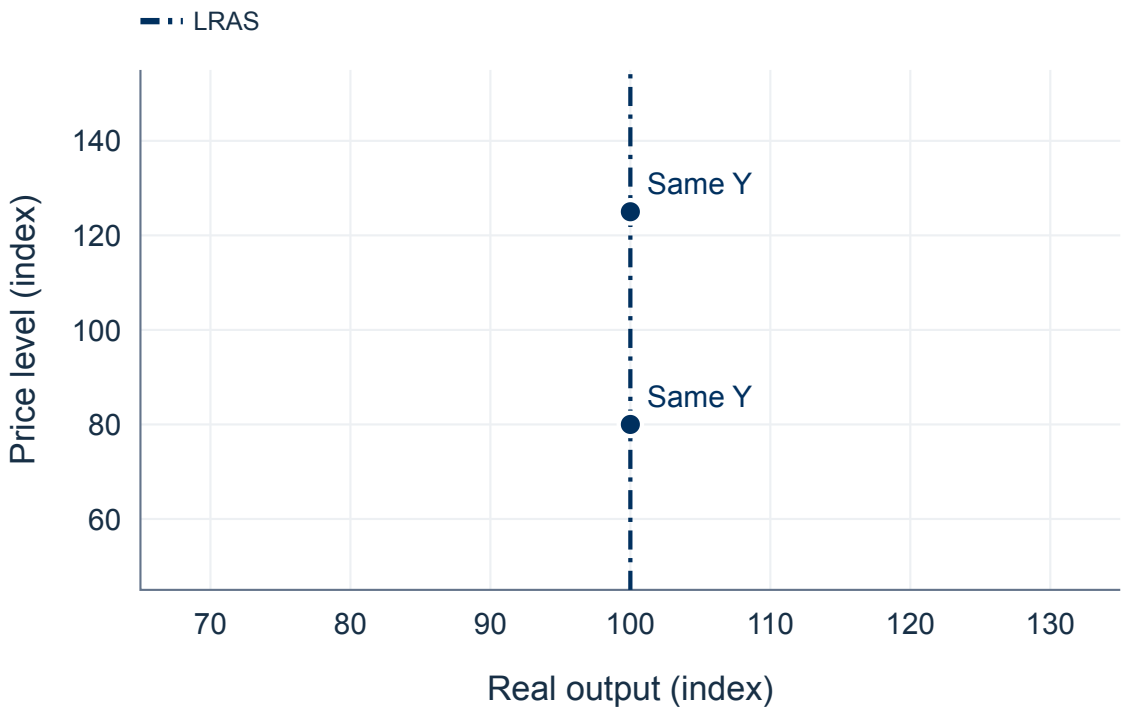
Sustainable production at normal resource-use rates.

Long-run adjustment

Wages and prices have time to adjust.

Long-Run Aggregate Supply

Potential output = 100 · Output and price level are indices



Potential output

Sustainable production at normal resource-use rates.

Long-run adjustment

Wages and prices have time to adjust.

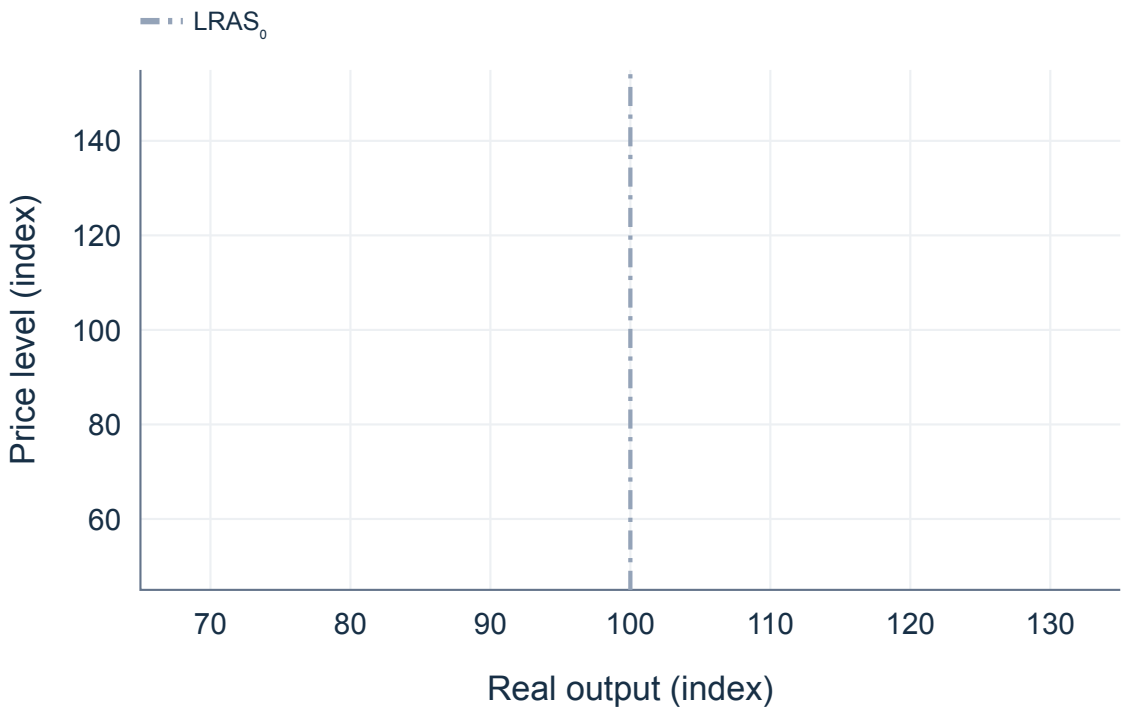
Vertical LRAS

The price level alone does not change productive capacity.

Capital, workers, skills, and technology determine potential output.

Factors That Shift LRAS

Separate changes from the same initial capacity

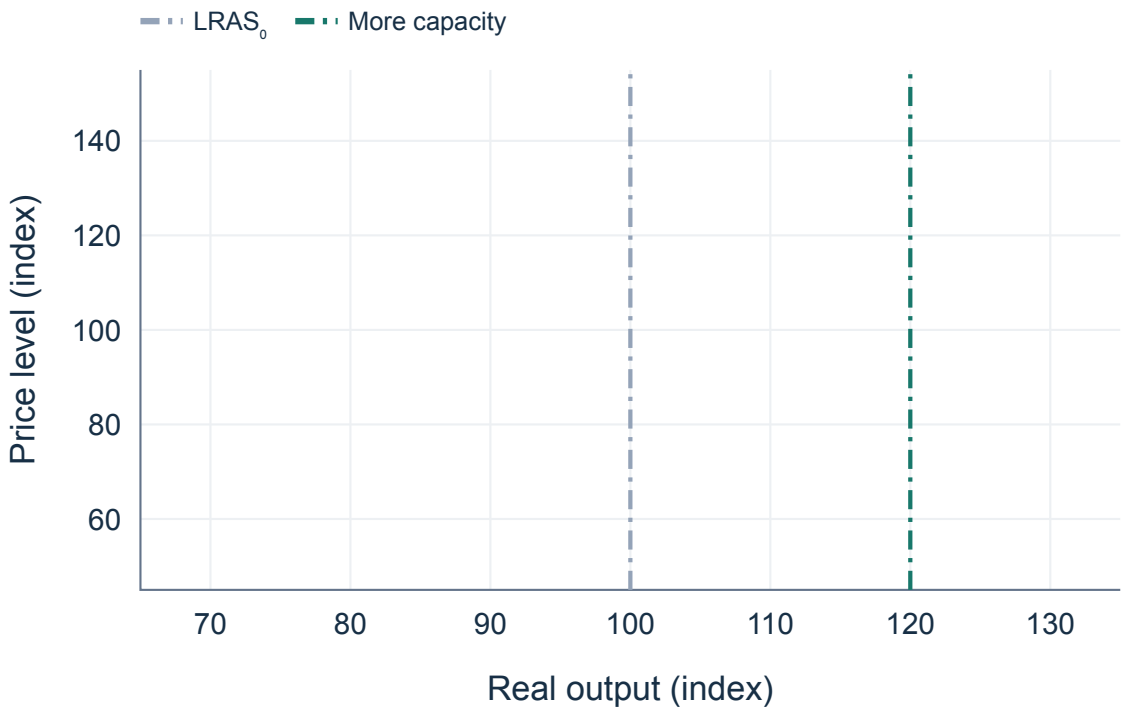


Starting capacity

Potential output is 100.

Factors That Shift LRAS

Separate changes from the same initial capacity



Starting capacity

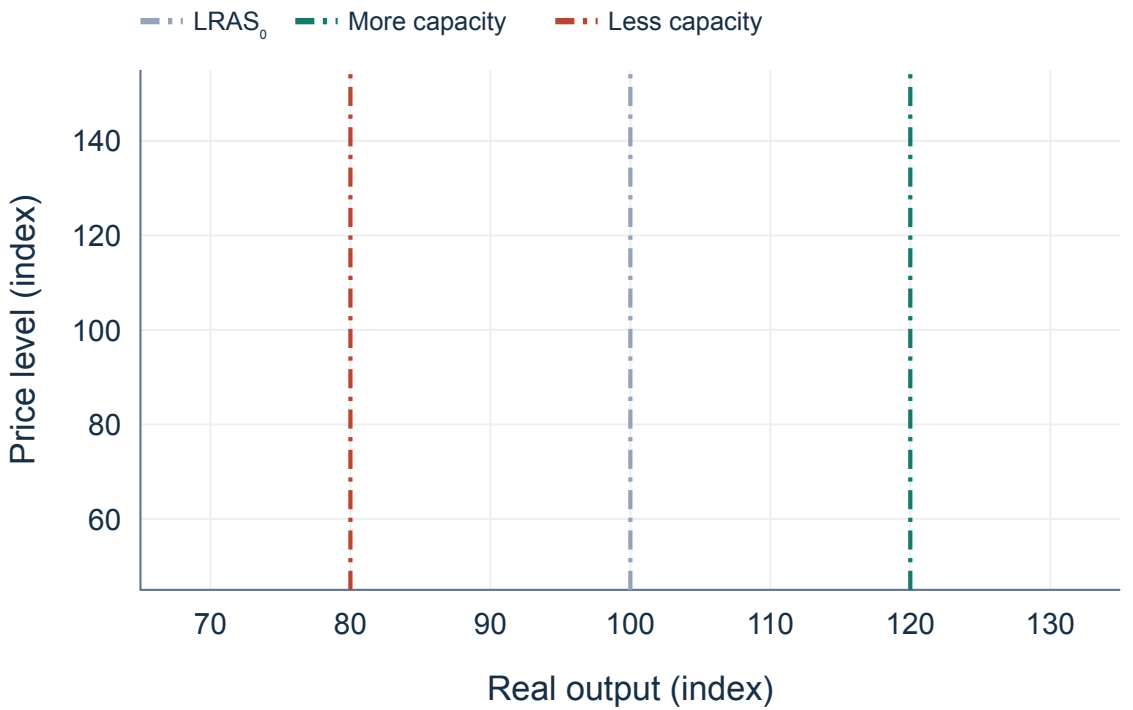
Potential output is 100.

More productive resources

Better technology or more capital shifts LRAS right.

Factors That Shift LRAS

Separate changes from the same initial capacity



Starting capacity

Potential output is 100.

More productive resources

Better technology or more capital shifts LRAS right.

Lost productive resources

Destroyed capital shifts LRAS left.

A price-level change alone does not shift LRAS.

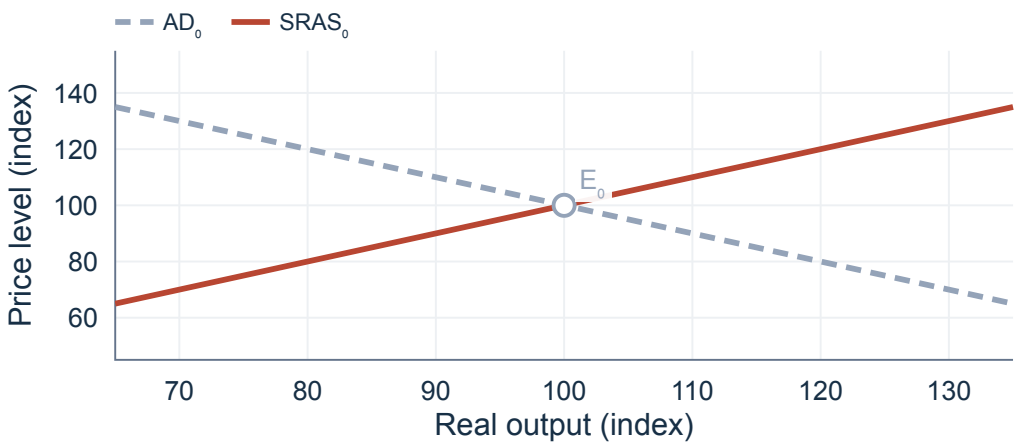
Short-Run Equilibrium: Demand and Supply Contractions

Above P_e: excess supply puts downward pressure on prices.

Below P_e: excess demand puts upward pressure on prices.

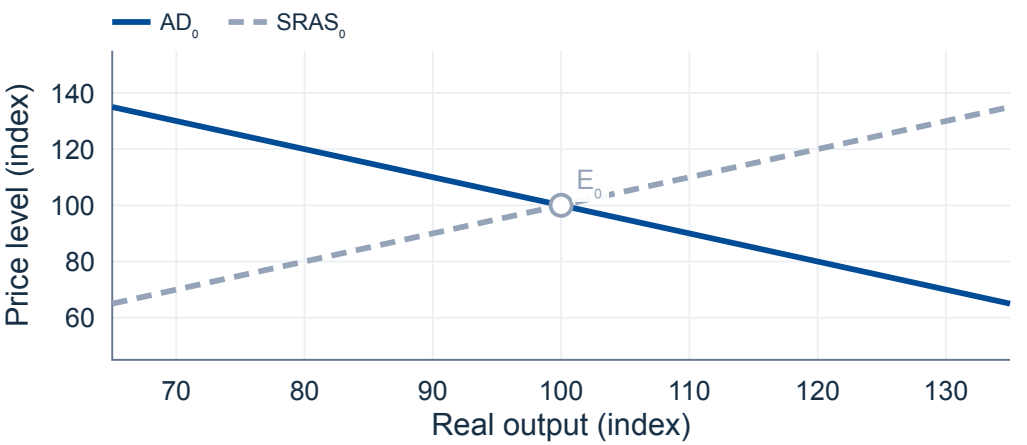
Demand contraction

Lower confidence reduces consumption.



Supply contraction

Higher oil prices raise production costs.



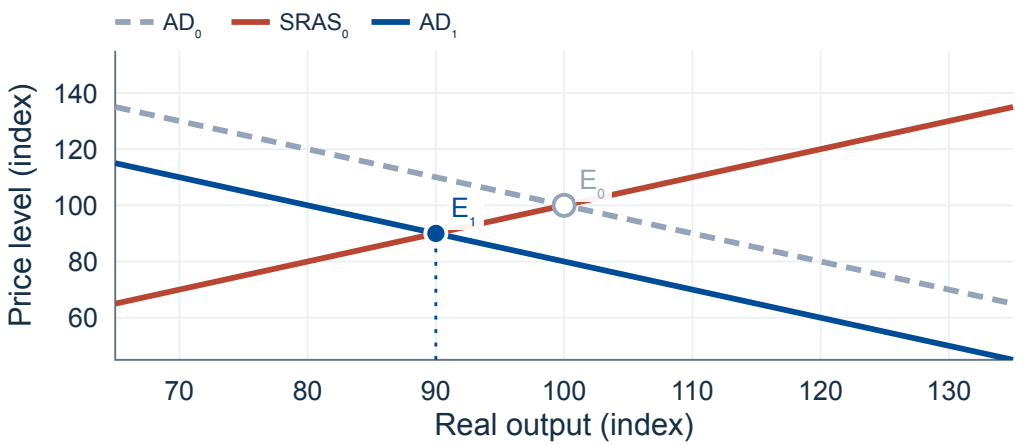
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Demand contraction

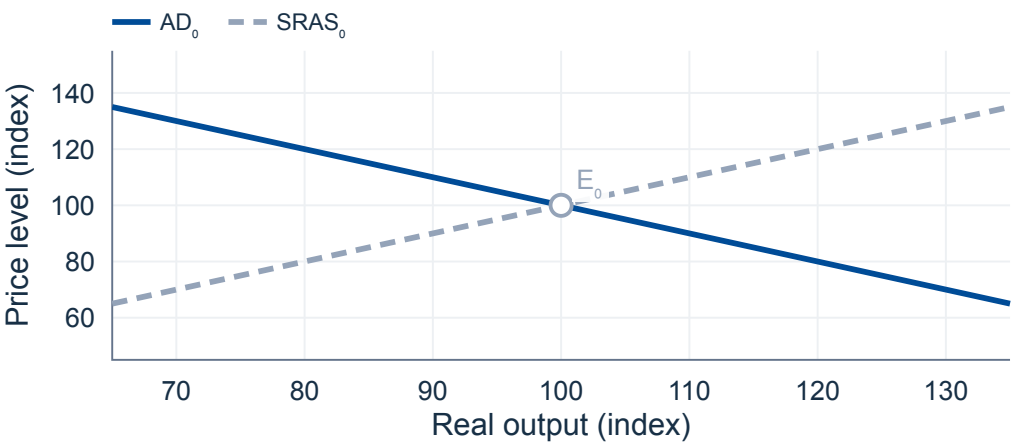
Lower confidence reduces consumption.



AD shifts left: the old price is above the new P_e.

Supply contraction

Higher oil prices raise production costs.



New equilibrium: Y falls; P falls.

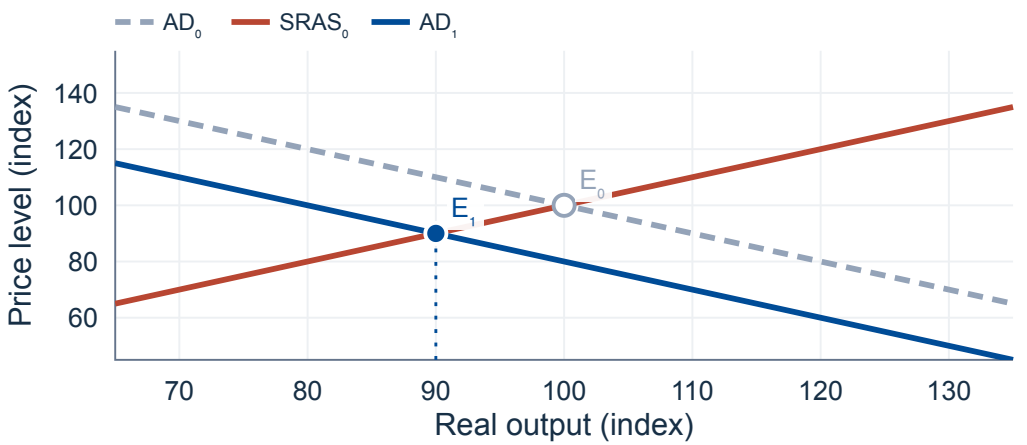
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Demand contraction

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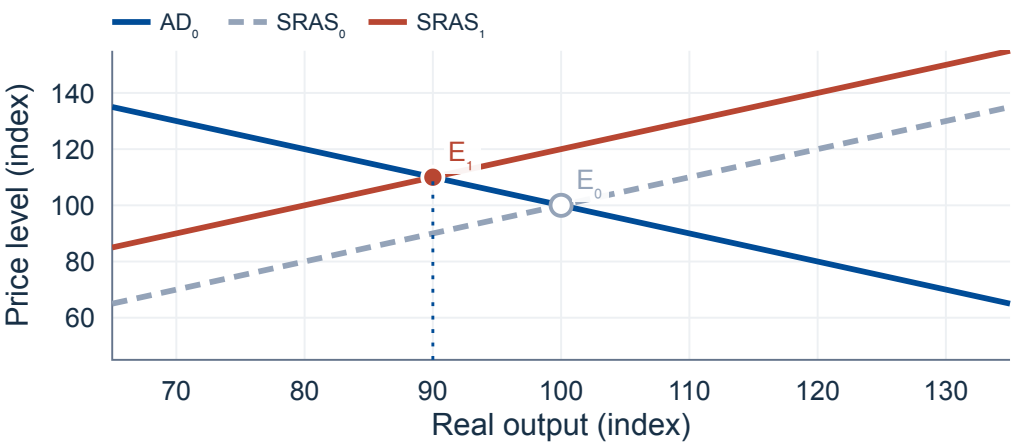


AD shifts left: the old price is above the new P_e.

New equilibrium: Y falls; P falls.

Supply contraction

Higher oil prices raise production costs.



SRAS shifts left: the old price is below the new P_e.

New equilibrium: Y falls; P rises.

Demand and Supply Shocks

Shock	Curve shift	Short-run outcome
Lower confidence		
Higher energy costs		

Demand and Supply Shocks

Shock	Curve shift	Short-run outcome
Lower confidence	AD shifts left	Y falls; P falls
Higher energy costs		

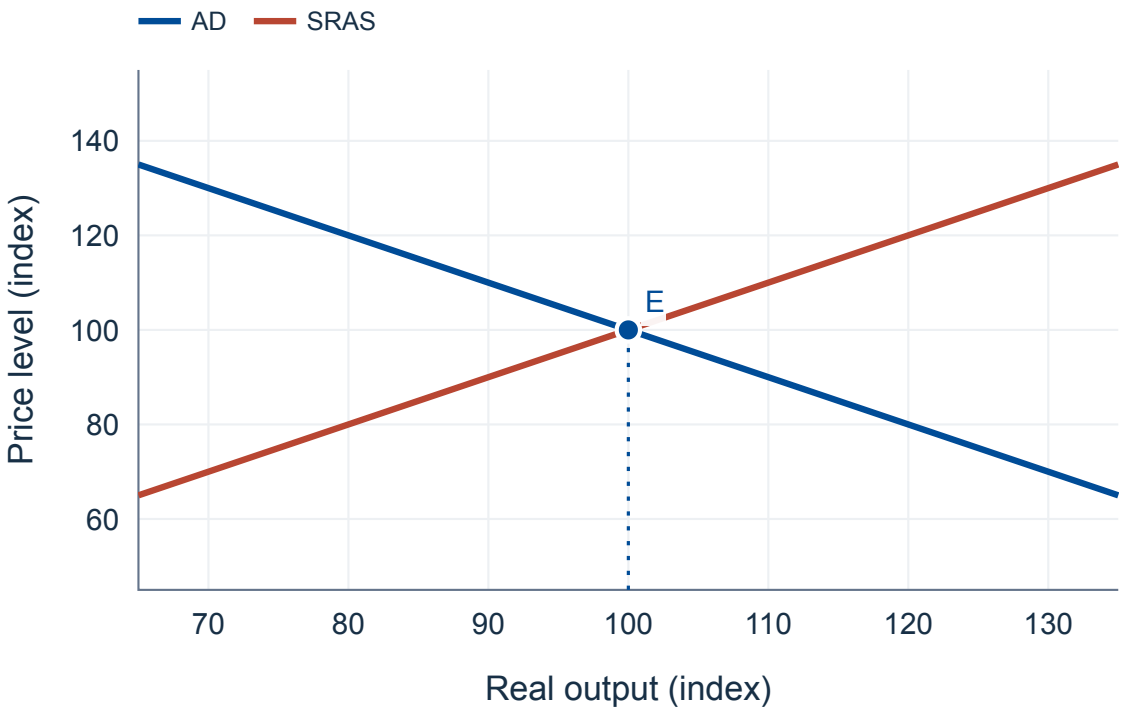
Demand and Supply Shocks

Shock	Curve shift	Short-run outcome
Lower confidence	AD shifts left	Y falls; P falls
Higher energy costs	SRAS shifts left	Y falls; P rises

Both can lower output, but their price-level effects differ.

Long-Run Macroeconomic Equilibrium

Potential output = 100

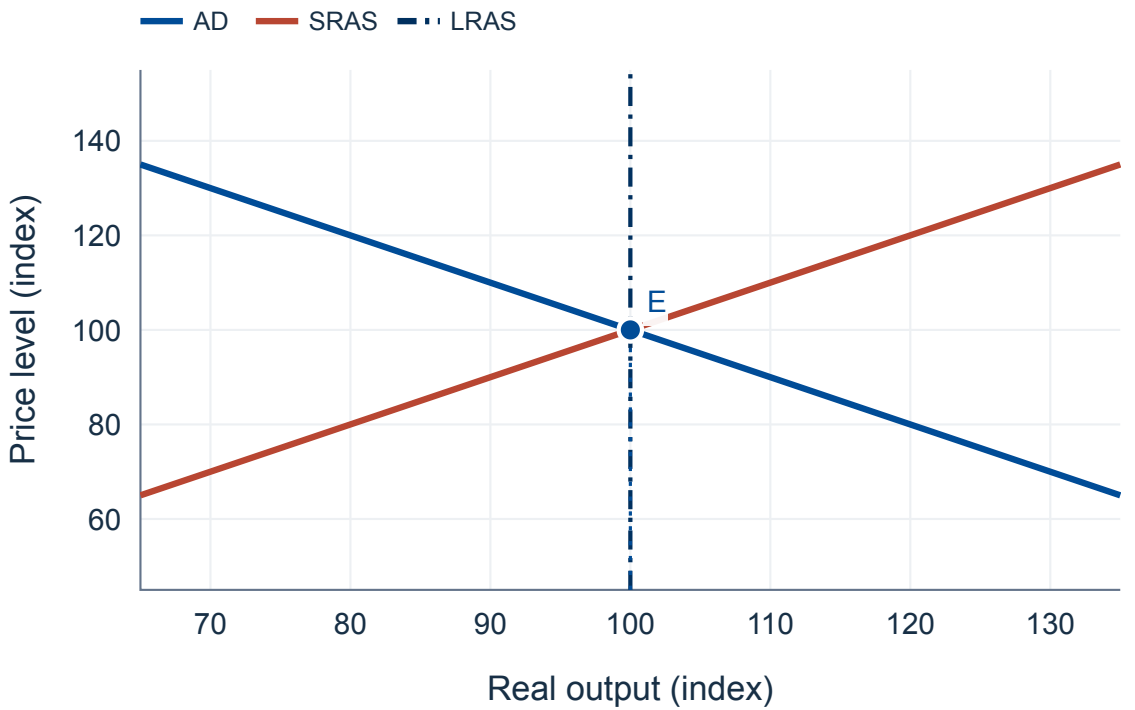


Short-run balance

AD and SRAS intersect.

Long-Run Macroeconomic Equilibrium

Potential output = 100



Short-run balance

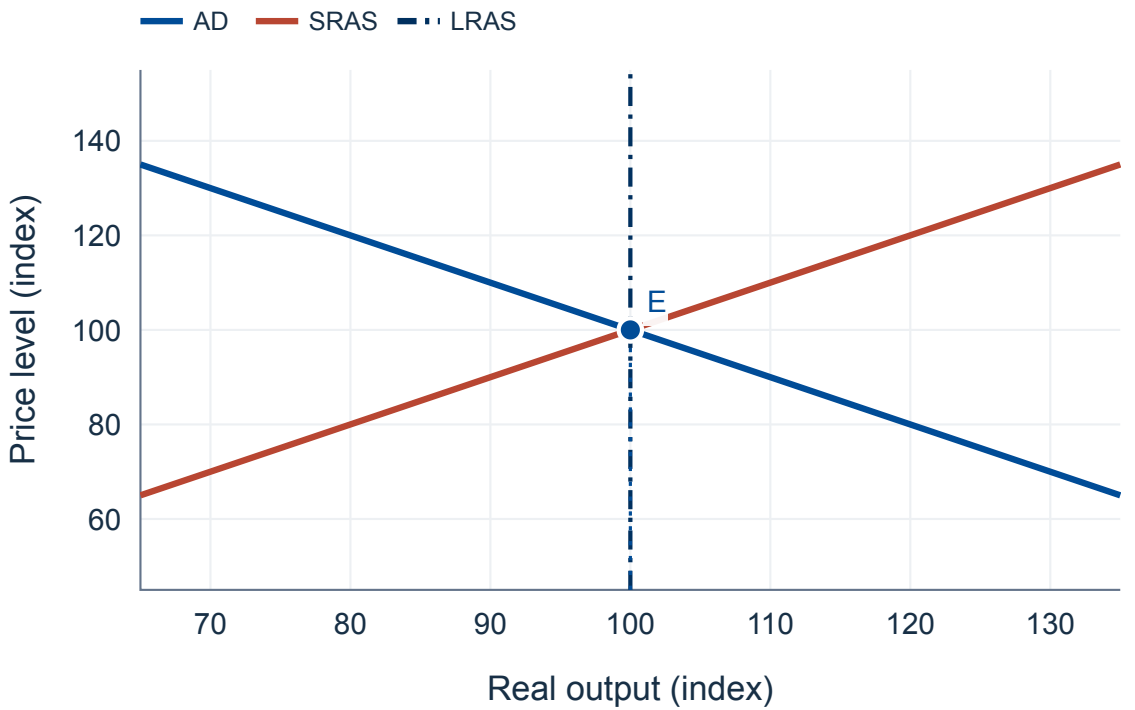
AD and SRAS intersect.

Capacity comparison

The intersection also lies on LRAS.

Long-Run Macroeconomic Equilibrium

Potential output = 100



Short-run balance

AD and SRAS intersect.

Capacity comparison

The intersection also lies on LRAS.

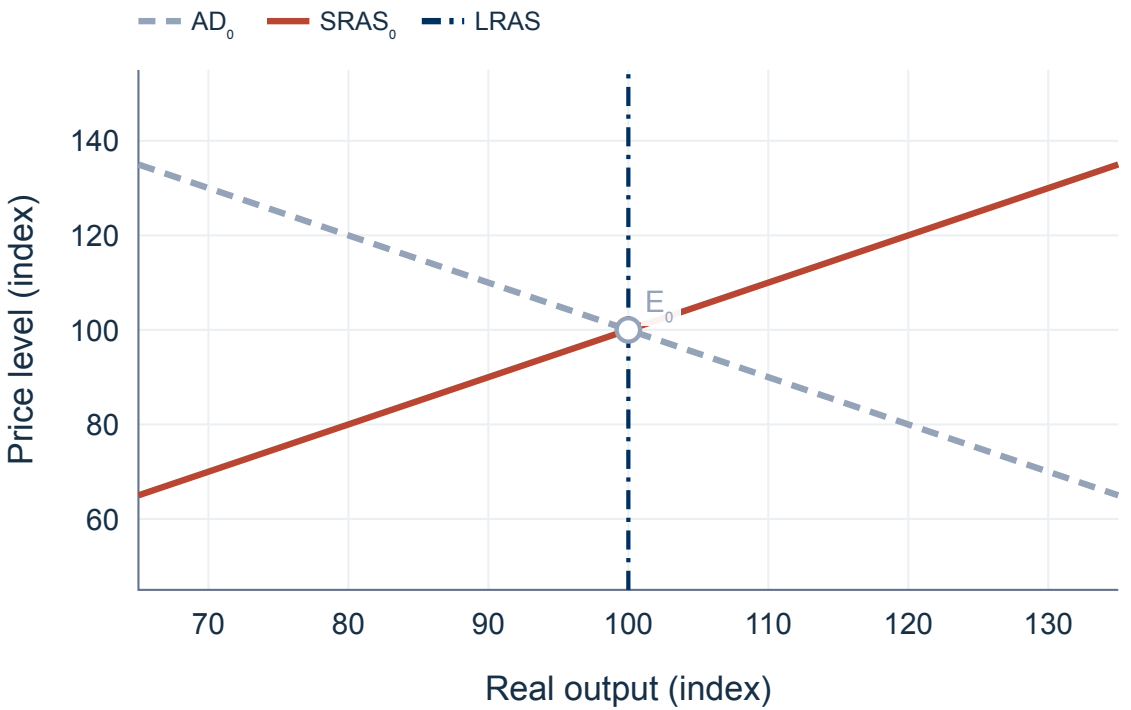
Long-run equilibrium

Actual output equals potential output.

Output gaps compare actual output with potential output.

Output Gaps

One shock: consumption falls · Potential output remains 100

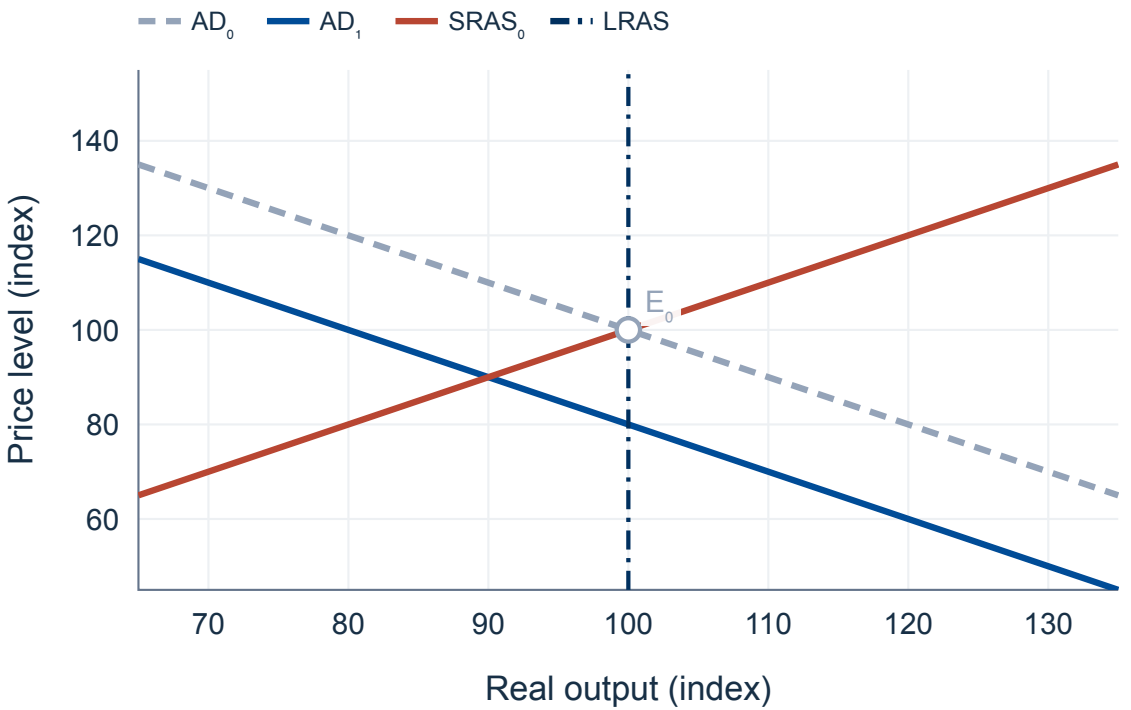


Initial equilibrium

Y = 100: output equals potential.

Output Gaps

One shock: consumption falls · Potential output remains 100



Initial equilibrium

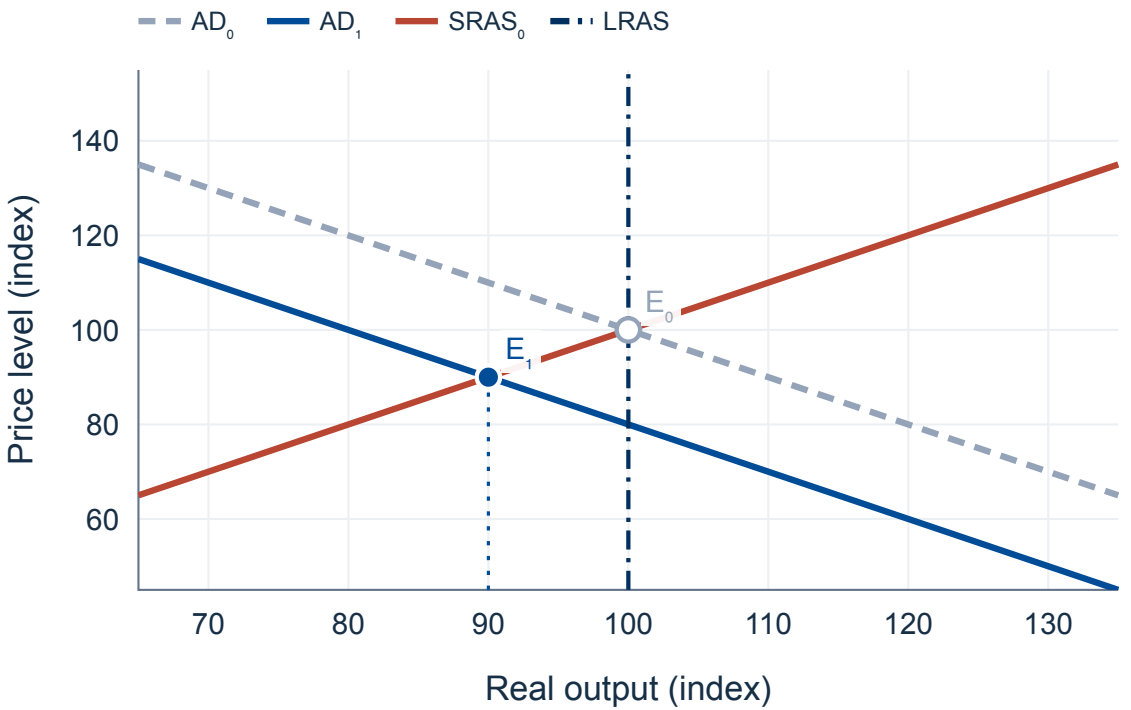
Y = 100: output equals potential.

Demand falls

AD shifts left; SRAS and LRAS remain fixed.

Output Gaps

One shock: consumption falls · Potential output remains 100



Initial equilibrium

Y = 100: output equals potential.

Demand falls

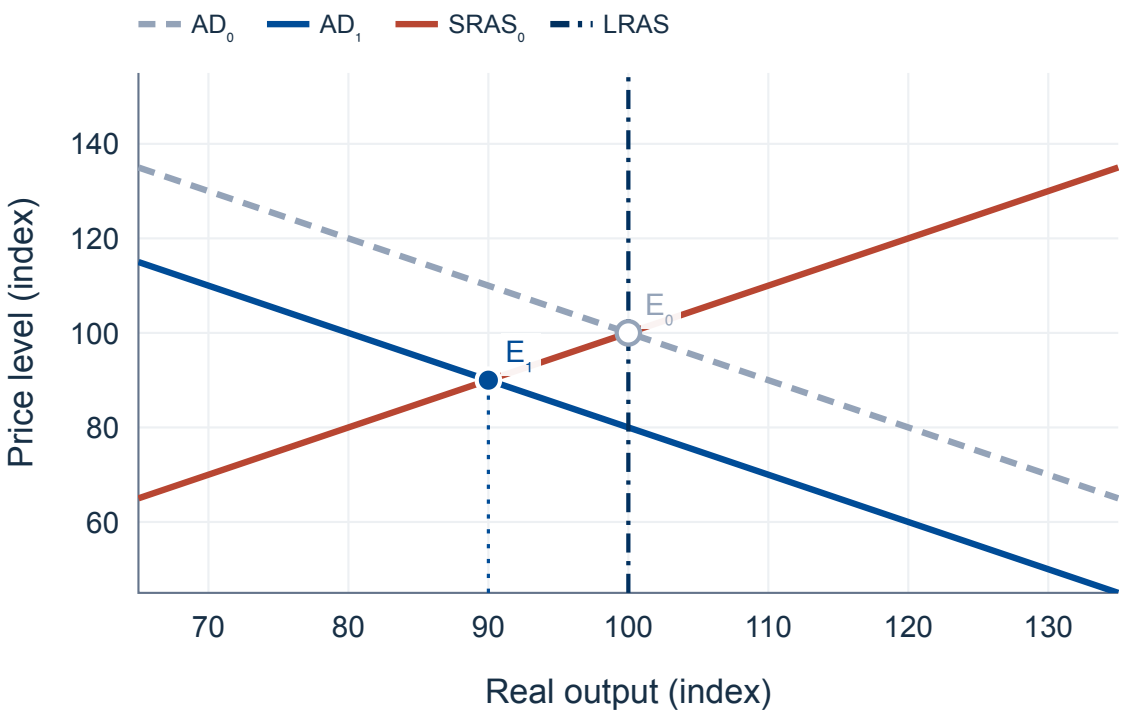
AD shifts left; SRAS and LRAS remain fixed.

Short-run equilibrium

Y = 90: a recessionary gap.

Output Gaps

One shock: consumption falls · Potential output remains 100



Initial equilibrium

Y = 100: output equals potential.

Demand falls

AD shifts left; SRAS and LRAS remain fixed.

Short-run equilibrium

Y = 90: a recessionary gap.

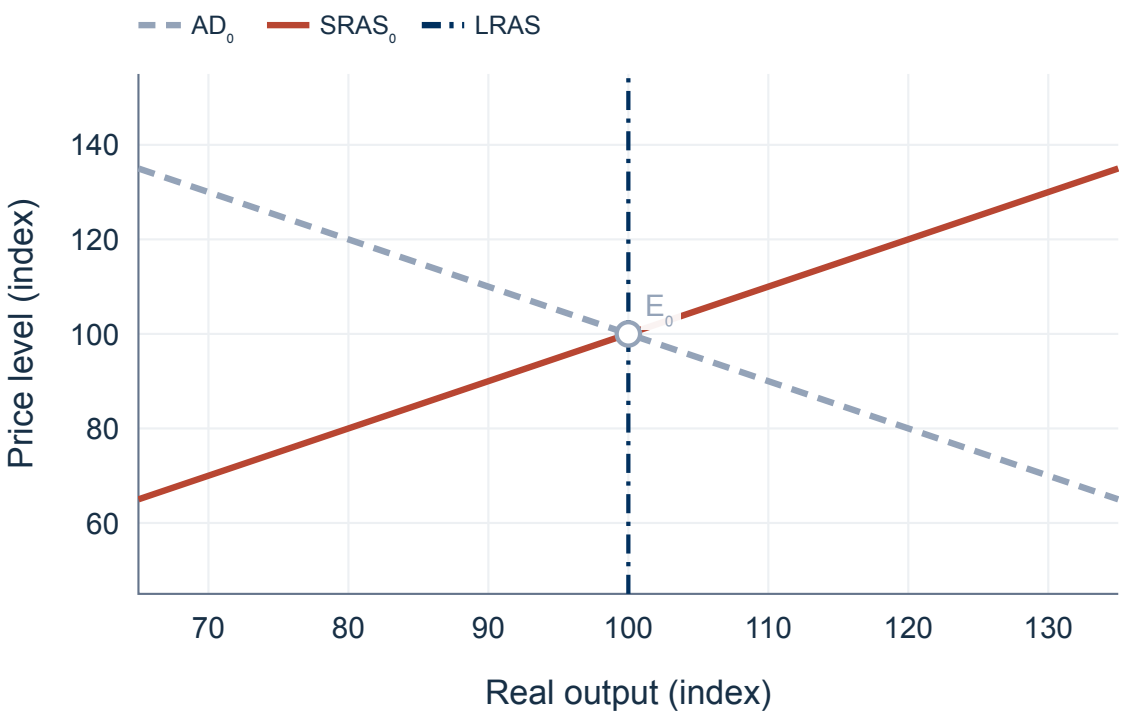
Measure the gap

(90 − 100) ÷ 100 × 100 = −10%.

Y < potential: recessionary gap. Y > potential: inflationary gap.

Adjustment Back to Potential

Demand remains lower · Productive capacity is unchanged

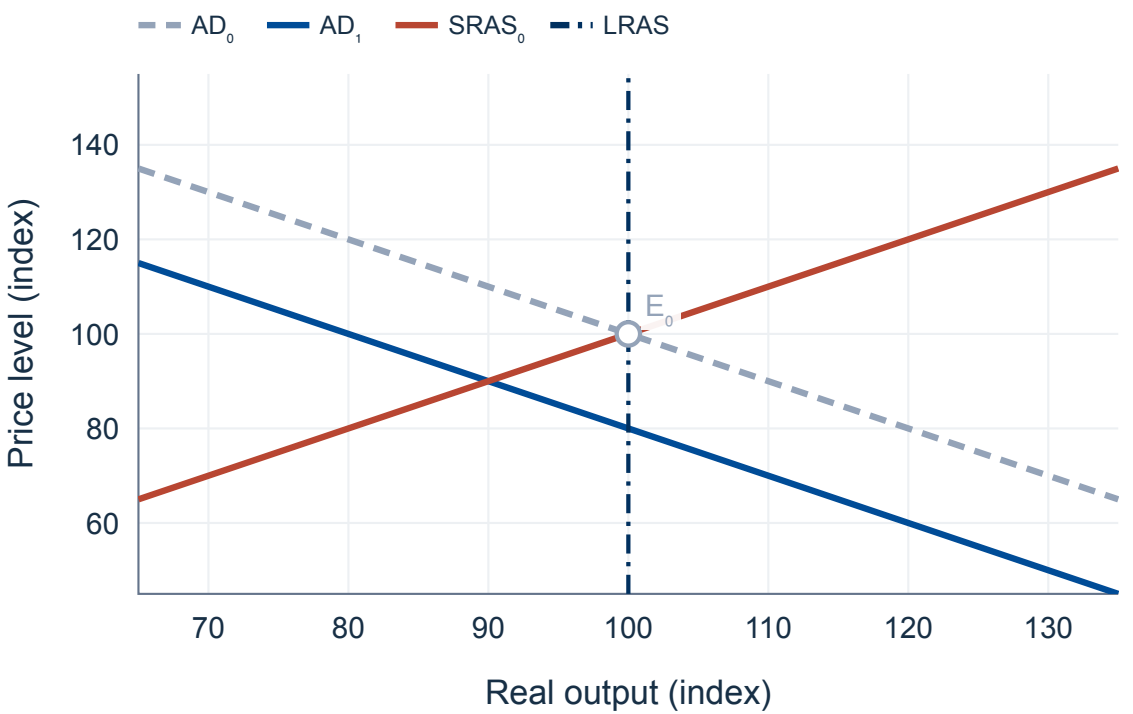


Before the shock

Output and potential both equal 100.

Adjustment Back to Potential

Demand remains lower · Productive capacity is unchanged

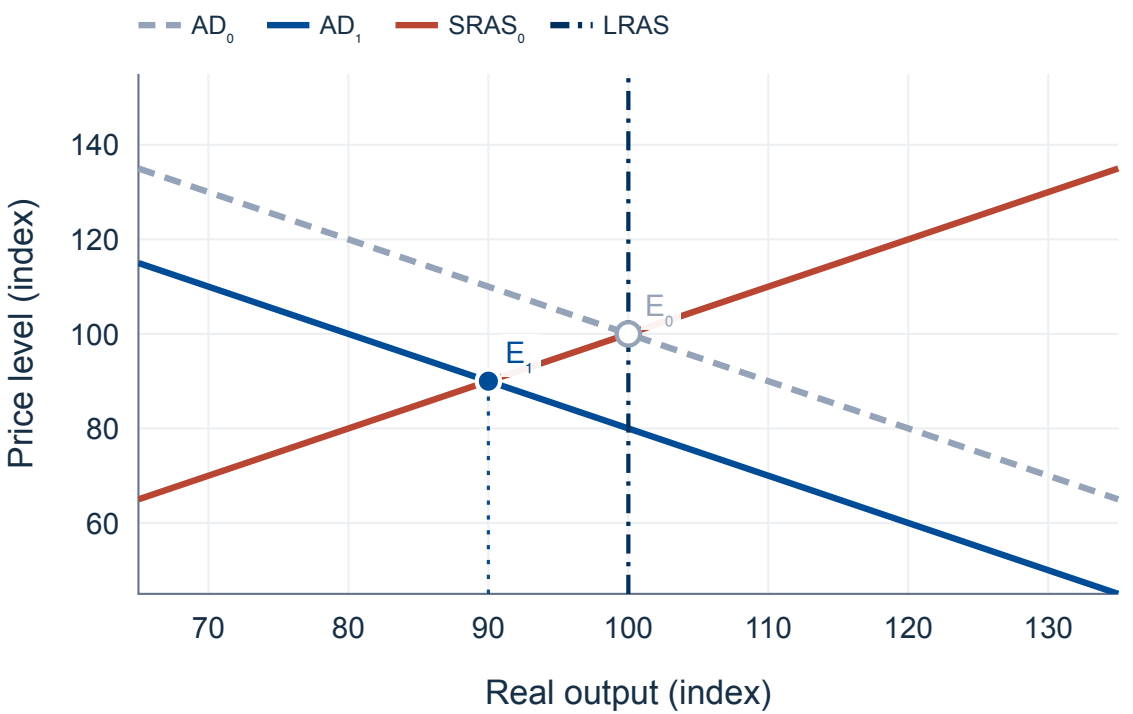


Before the shock
Output and potential both equal 100.

Demand falls
AD shifts left.

Adjustment Back to Potential

Demand remains lower · Productive capacity is unchanged



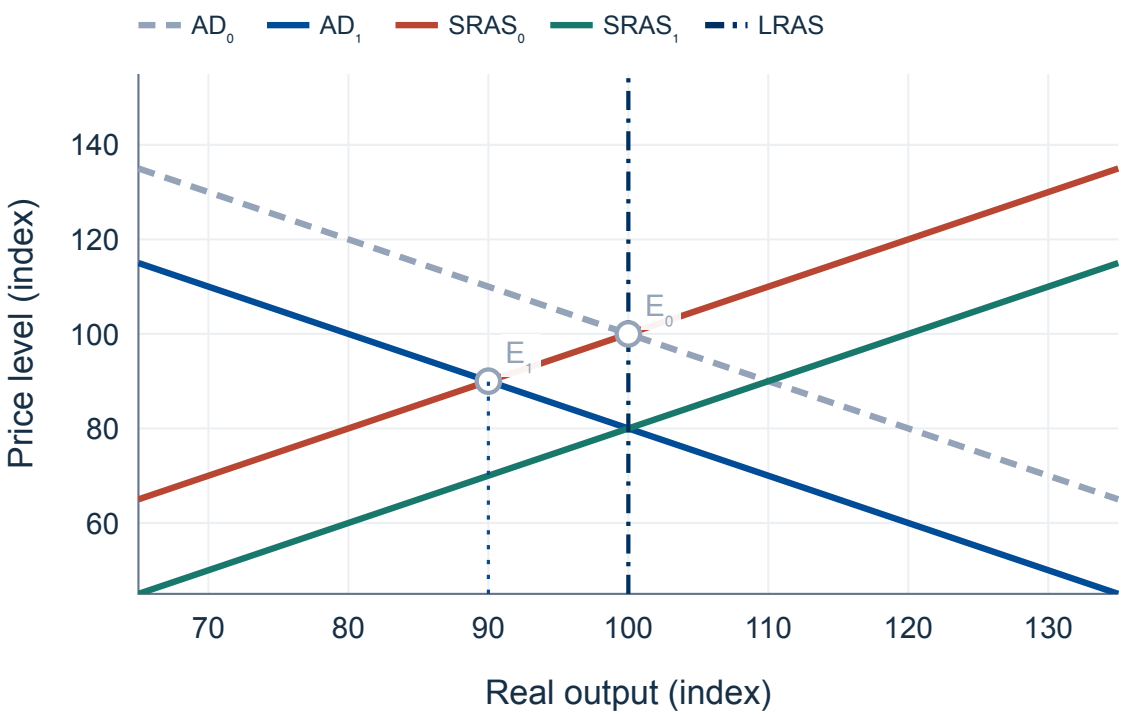
Before the shock
Output and potential both equal 100.

Demand falls
AD shifts left.

Short-run outcome
Y = 90 and P = 90.

Adjustment Back to Potential

Demand remains lower · Productive capacity is unchanged



Before the shock
Output and potential both equal 100.

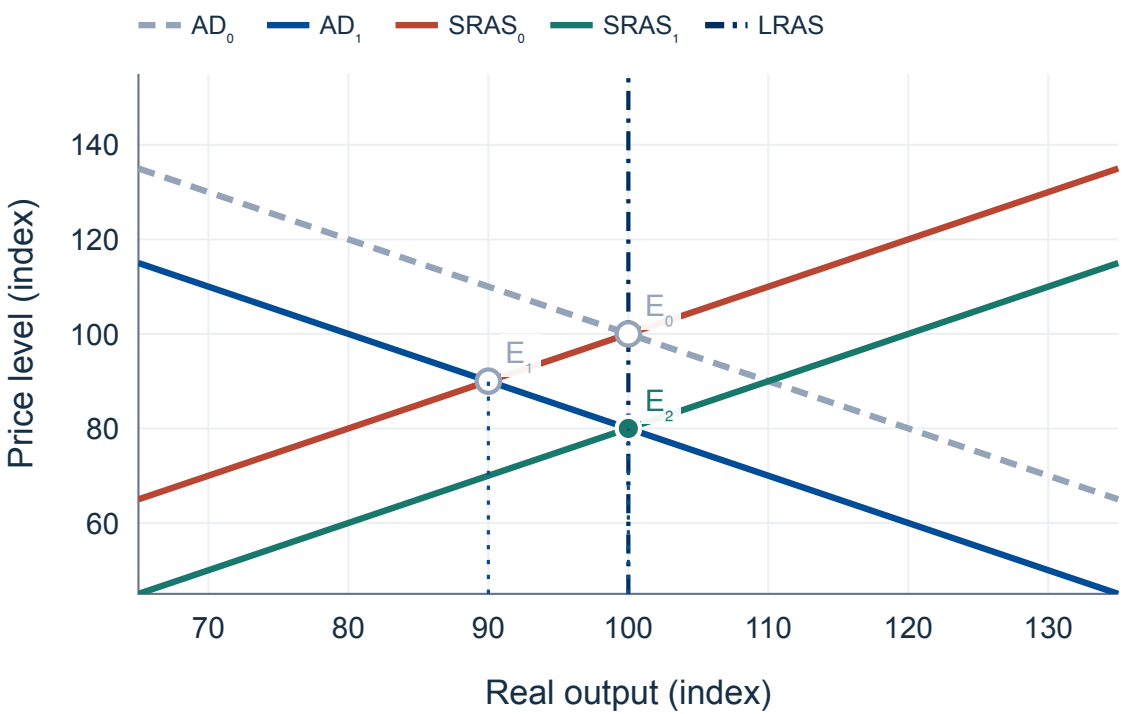
Demand falls
AD shifts left.

Short-run outcome
Y = 90 and P = 90.

Wages and costs adjust
Spare capacity gradually shifts SRAS right.

Adjustment Back to Potential

Demand remains lower · Productive capacity is unchanged



Before the shock
Output and potential both equal 100.

Demand falls
AD shifts left.

Short-run outcome
Y = 90 and P = 90.

Wages and costs adjust
Spare capacity gradually shifts SRAS right.

Long-run outcome
Y returns to 100; P falls to 80.

AD and LRAS stay fixed during the wage-and-cost adjustment.

Output Gap Checkpoint

Calculate each output gap. Name the gap and the direction of SRAS adjustment.

A. Potential GDP = $500 billion; actual GDP = $450 billion.

B. Potential GDP = $500 billion; actual GDP = $525 billion.

Output Gap Checkpoint Answers

Case	Output gap	Cost adjustment
A: 450 versus 500		
B: 525 versus 500		

Output Gap Checkpoint Answers

Case	Output gap	Cost adjustment
A: 450 versus 500	−10%: recessionary	Lower cost pressure; SRAS right
B: 525 versus 500		

Output Gap Checkpoint Answers

Case	Output gap	Cost adjustment
A: 450 versus 500	−10%: recessionary	Lower cost pressure; SRAS right
B: 525 versus 500	+5%: inflationary	Higher cost pressure; SRAS left

Hold AD and potential output fixed during adjustment.

Appendix: AD and Income–Expenditure

> **At a given price level**
>
> The income–expenditure model finds output where AE = Y.

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Compare a lower price level

Purchasing power and interest-sensitive spending can rise.

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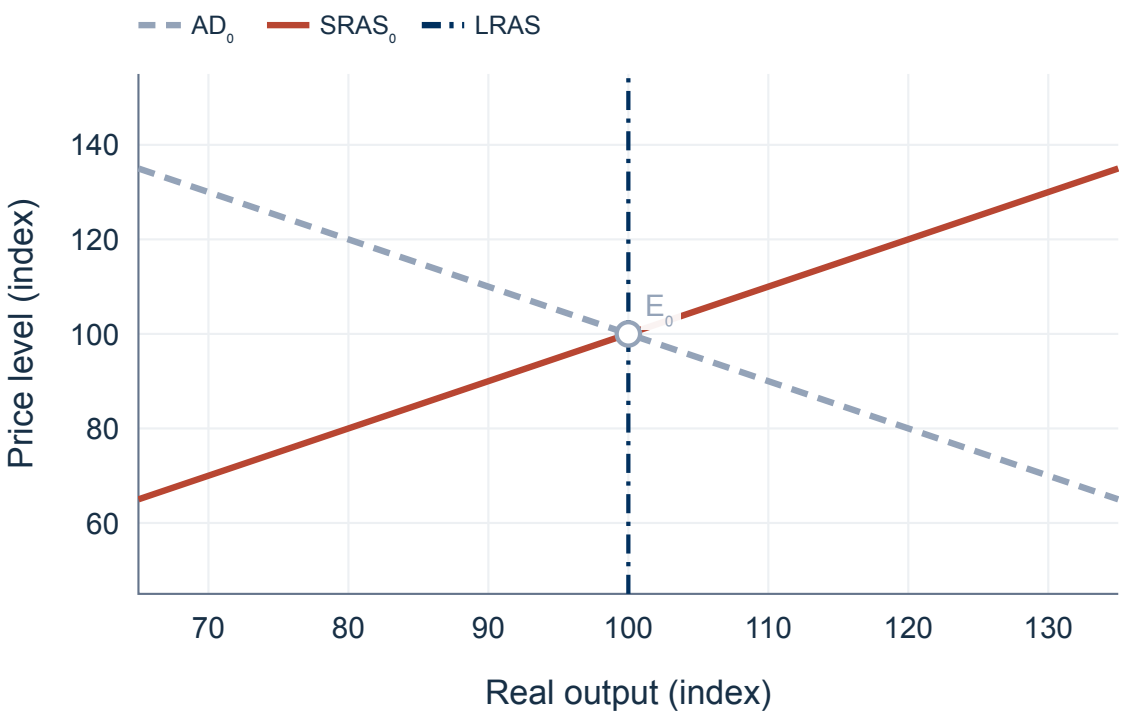
Trace aggregate demand

Higher planned spending produces a larger equilibrium Y.

Changing the price level traces AD; changing other demand influences shifts AD.

Worked Case: Lower Consumption

Initial Y = 100 · Initial P = 100 · Potential output = 100

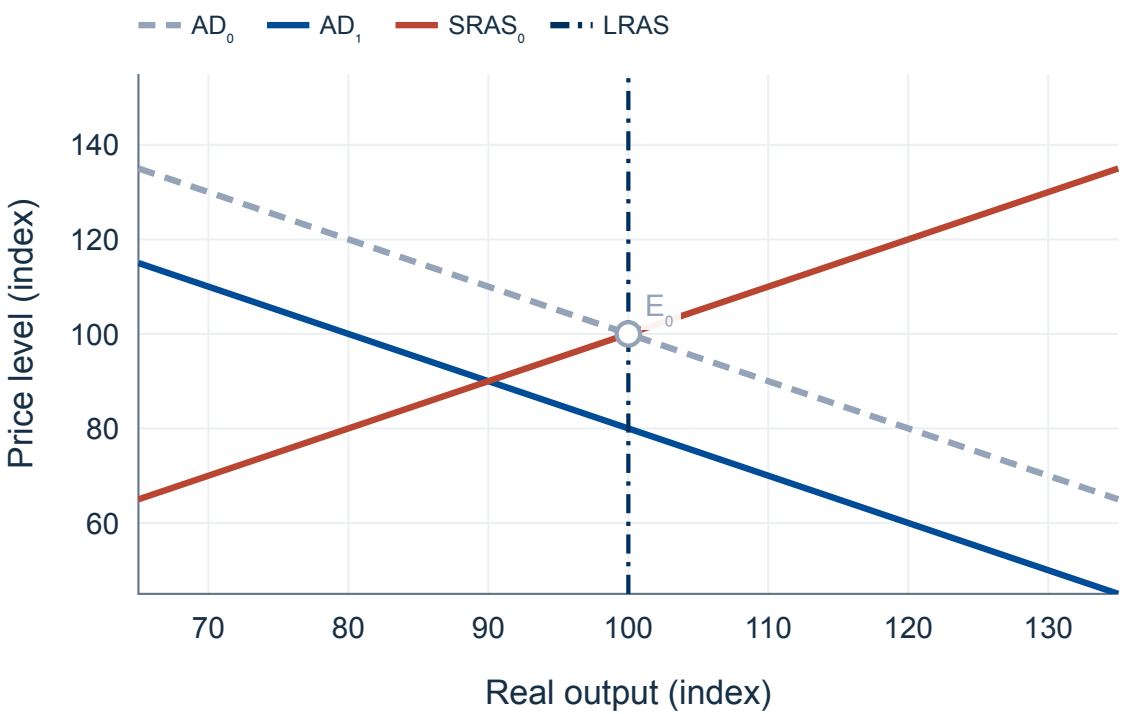


Initial equilibrium

AD, SRAS, and LRAS meet at potential.

Worked Case: Lower Consumption

Initial Y = 100 · Initial P = 100 · Potential output = 100

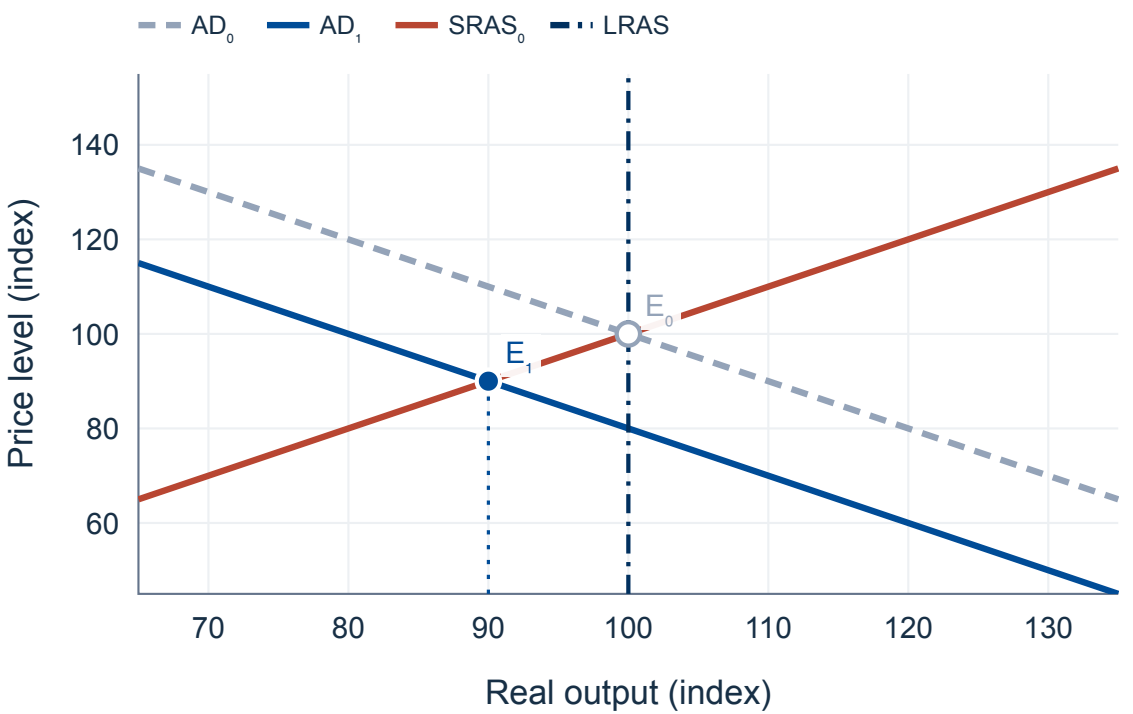


Initial equilibrium
AD, SRAS, and LRAS meet at potential.

Initial shock
Consumption falls; AD shifts left.

Worked Case: Lower Consumption

Initial Y = 100 · Initial P = 100 · Potential output = 100



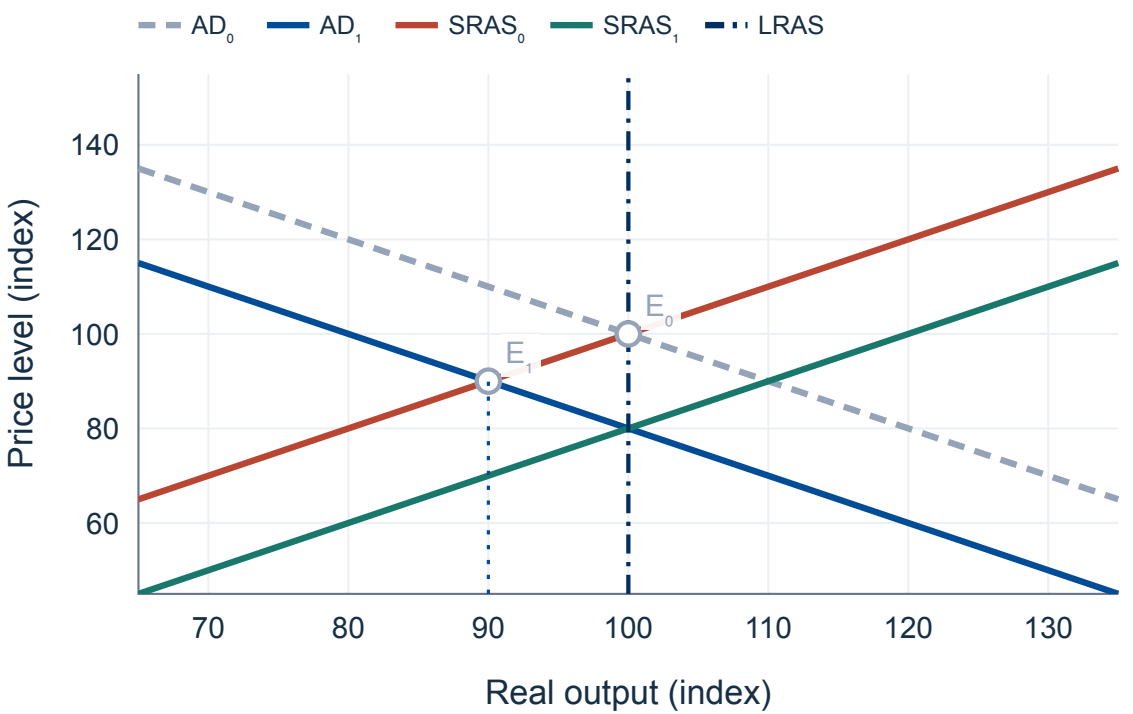
Initial equilibrium
AD, SRAS, and LRAS meet at potential.

Initial shock
Consumption falls; AD shifts left.

Short-run equilibrium
Y = 90 and P = 90: a recessionary gap.

Worked Case: Lower Consumption

Initial Y = 100 · Initial P = 100 · Potential output = 100



Initial equilibrium
AD, SRAS, and LRAS meet at potential.

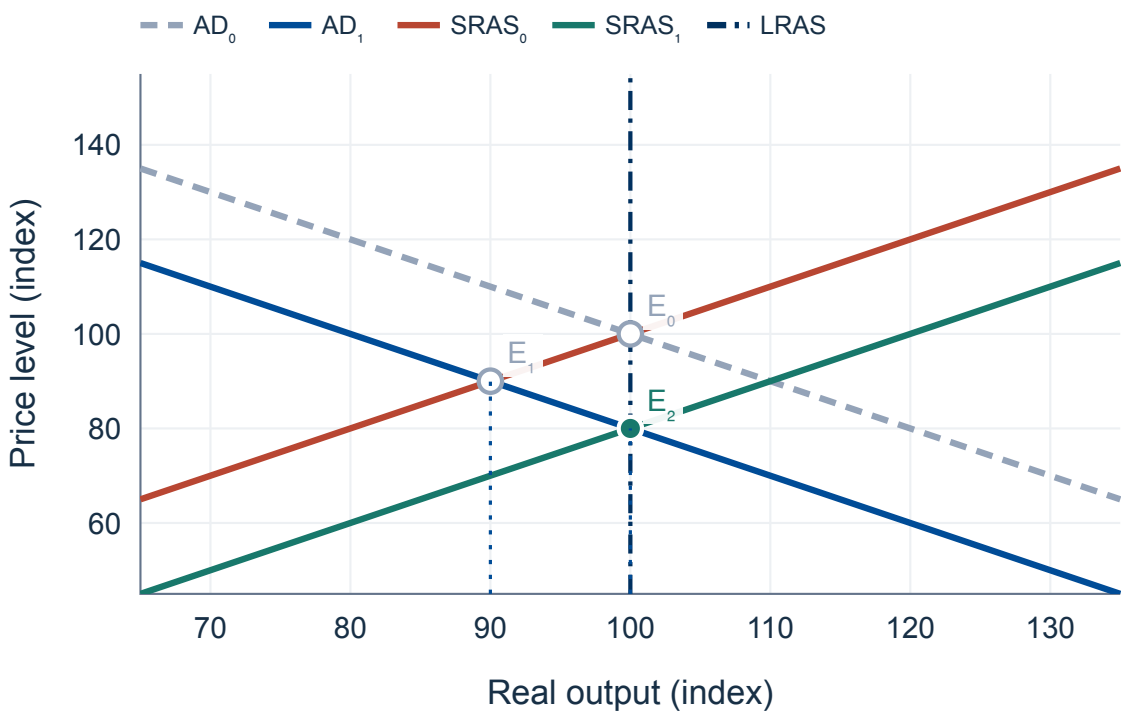
Initial shock
Consumption falls; AD shifts left.

Short-run equilibrium
Y = 90 and P = 90: a recessionary gap.

Later adjustment
Lower cost pressure shifts SRAS right.

Worked Case: Lower Consumption

Initial Y = 100 · Initial P = 100 · Potential output = 100



Initial equilibrium
AD, SRAS, and LRAS meet at potential.

Initial shock
Consumption falls; AD shifts left.

Short-run equilibrium
Y = 90 and P = 90: a recessionary gap.

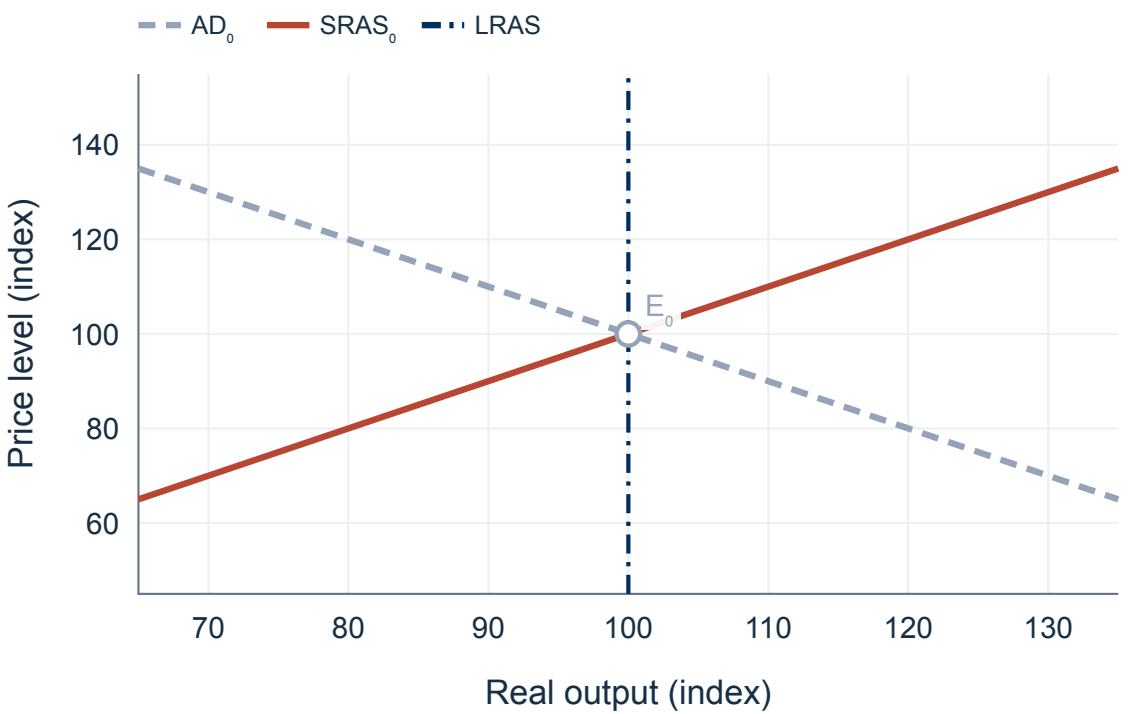
Later adjustment
Lower cost pressure shifts SRAS right.

Final equilibrium
Y = 100 and P = 80: potential is restored.

Identify the shock, shift one curve, locate equilibrium, then explain adjustment.

Worked Case: Higher Investment

Initial Y = 100 · Initial P = 100 · Potential output = 100

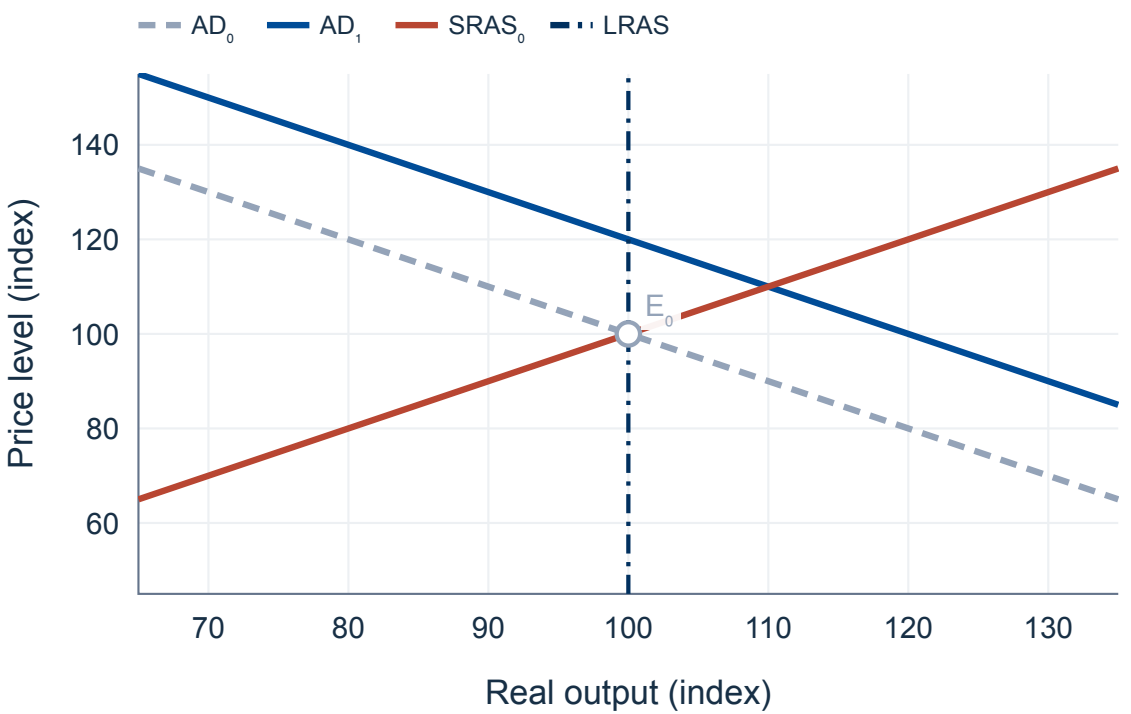


Initial equilibrium

AD, SRAS, and LRAS meet at potential.

Worked Case: Higher Investment

Initial Y = 100 · Initial P = 100 · Potential output = 100

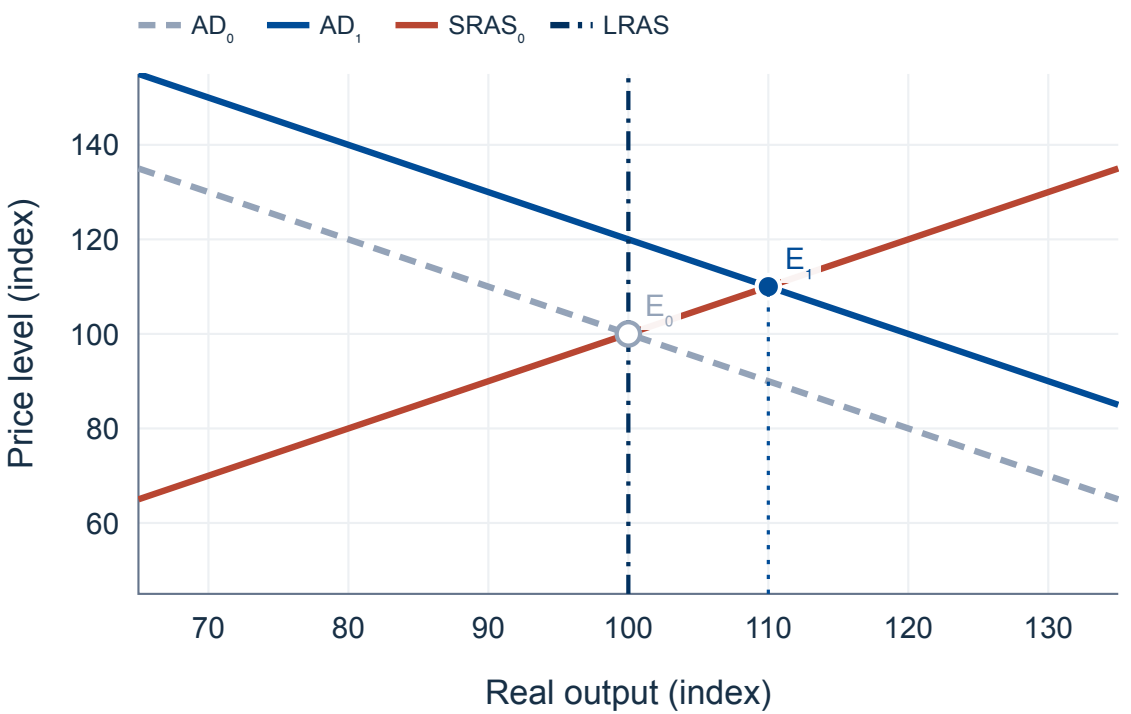


Initial equilibrium
AD, SRAS, and LRAS meet at potential.

Initial shock
Investment rises; AD shifts right.

Worked Case: Higher Investment

Initial Y = 100 · Initial P = 100 · Potential output = 100



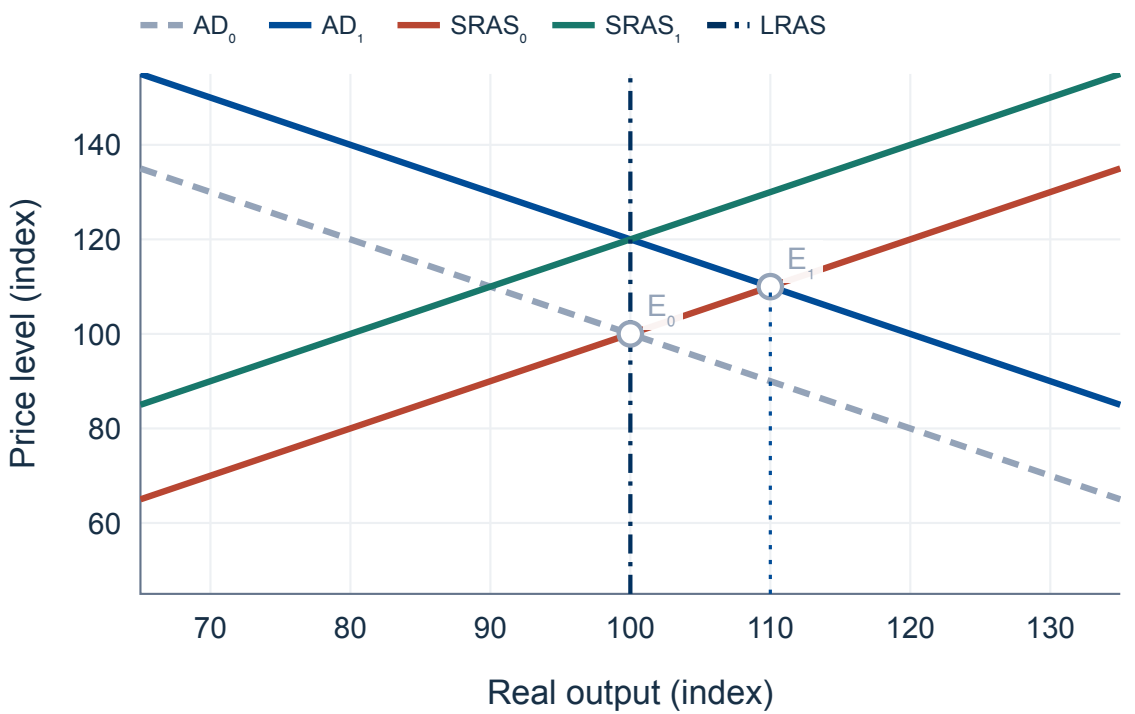
Initial equilibrium
AD, SRAS, and LRAS meet at potential.

Initial shock
Investment rises; AD shifts right.

Short-run equilibrium
Y = 110 and P = 110: an inflationary gap.

Worked Case: Higher Investment

Initial Y = 100 · Initial P = 100 · Potential output = 100



Initial equilibrium
AD, SRAS, and LRAS meet at potential.

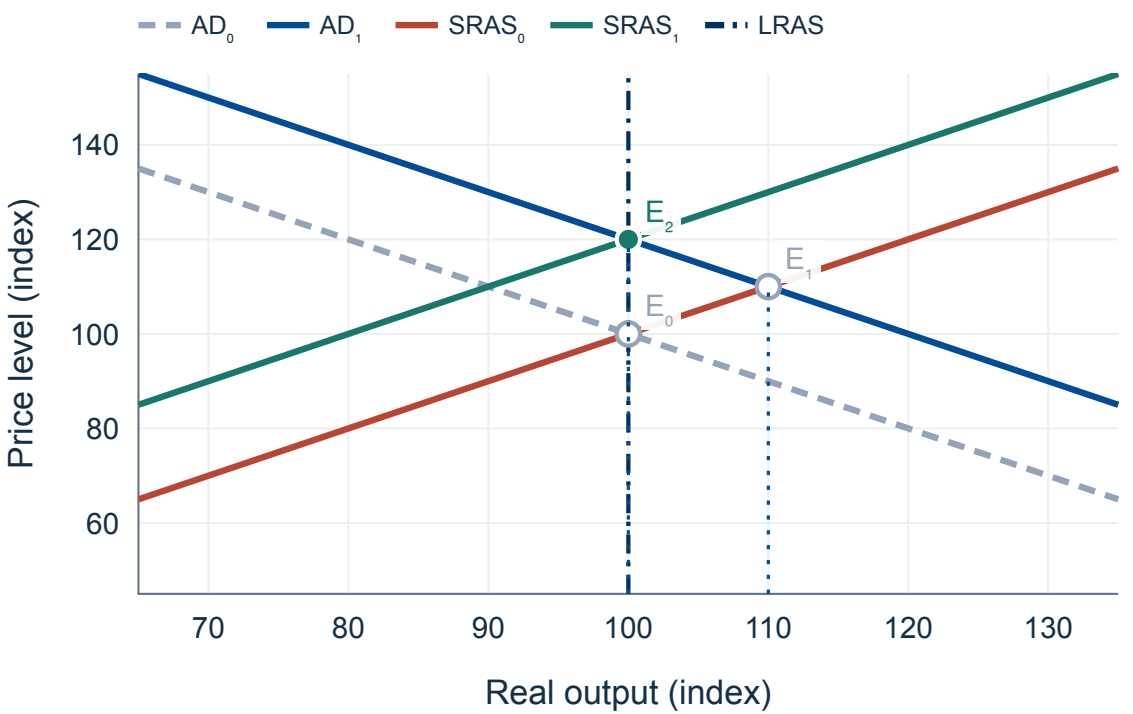
Initial shock
Investment rises; AD shifts right.

Short-run equilibrium
Y = 110 and P = 110: an inflationary gap.

Later adjustment
Higher wages and costs shift SRAS left.

Worked Case: Higher Investment

Initial Y = 100 · Initial P = 100 · Potential output = 100



Initial equilibrium
AD, SRAS, and LRAS meet at potential.

Initial shock
Investment rises; AD shifts right.

Short-run equilibrium
Y = 110 and P = 110: an inflationary gap.

Later adjustment
Higher wages and costs shift SRAS left.

Final equilibrium
Y = 100 and P = 120: potential is restored.

Identify the shock, shift one curve, locate equilibrium, then explain adjustment.